Path Planning of Autonomous Mobile Robot in Comprehensive Unknown Environment Using Deep Reinforcement Learning

Zekun Bai, Hui Pang, *Member, IEEE*, Zhaonian He, Bin Zhao, and Tong Wang

***Abstract*—In real-world situations, some unavoidable problems like significant dependence on environment information, long inference time and weak anti-disturbance ability are often involved in path planning of autonomous mobile robot (AMR) under unknown environments. To solve these issues, this article proposes an improved deep reinforcement learning-based path-planning algorithm to find out an optimized path for a class of AMRs. First, the path planning of AMR is described as a Markov decision process framework, and the double deep Q network (DDQN) is utilized to obtain the optimal adaptive solutions of AMR's path planning. Second, a comprehensive reward function integrated with heuristic function is designed to navigate the AMR into the target area. Afterwards, an optimized deep neural network with an adaptive ε-greedy action selection policy is designed to deal with the tradeoff between exploration and exploitation, thus further to improve the global searching capability and the convergence performance for the AMR path planning. Moreover, Bezier curve theory is utilized to smooth the planned path. Finally, the comparative simulations are carried out to validate our proposed path-planning algorithm. The results show that, compared with DQN, A*, RRT, and APF algorithms, our improved DDQN algorithm can produce safer and shorter global paths in comprehensive unknown environments. Meanwhile, the IDDQN algorithm has strong adaptability to random disturbances in unknown environments.**

***Index Terms*—Autonomous mobile robot (AMR), deep reinforcement learning (DRL), double deep Q network (DDQN), path planning, path smooth.**

I. Introduction

RECENTLY, autonomous mobile robot (AMR) has gained extensive application prospects in the fields of biomedicine, space exploration, engineering equipment overhaul and maintenance due to its merits, such as small size, flexible deployment, and independent completion of missions [1]. Path planning is one of the core functions of AMR fields. Hence, it is particularly important for researchers to improve the autonomous path-planning ability of AMR. Path-planning system needs to generate a feasible path while satisfying multiple constraints: as maintaining a safe distance from obstacles, having the shortest possible distance to the target, path smoothness, and environmental disturbance information [2], [3]. When performing a task in an unknown environment, AMRs should calculate and modify its feasible collision-free path based on the information gathered from the surroundings [4]. In addition, AMRs need to generate path navigation information that conforms to the constraints in the presence of potential interference from external environmental factors. Therefore, path-planning methods with a high degree of autonomy and learning capability are essential for the AMRs.

Considerable efforts have been reported on path-planning algorithms for robot system, which can be broadly divided into three categories: 1) traditional algorithms; 2) bionic algorithms; and 3) learning-based algorithms, as shown in Fig. 1. The traditional algorithms involve graph-based, sampling-based and data-based methods. Bionic algorithms include genetic algorithm, artificial colony algorithm, fuzzy-logic algorithm, etc. Learning-based algorithm includes deep reinforcement learning (DRL) algorithm and meta learning, etc. In detail, commonly used traditional methods include A-star algorithm (A*) [5], artificial potential field algorithm (APF) [6] and rapidly exploring random trees algorithm (RRT) [7]. Although A* algorithm can efficiently determine the shortest path, its computation will increase exponentially with the expansion of space, so it is unsuitable for complicated continuous obstacle environments [8]. The APF algorithm uses the potential function descent method to identify the ideal path [9]. Lin et al. [10] designed a two-level path-planning method, in which both modified APF algorithm and dynamic window algorithm were adopted to guide the robot to avoid obstacles and plan a shorter and smoother path reasonably. However, as the number of obstacles increases, APF algorithm will encounter an increasing number of zero-potential energy points in the environment, it is easy to trap in local minimum point and could not plan a complete path. RRT algorithm is a path-planning algorithm based on random sampling, although RRT algorithm has strong search ability and rapid search speed, it has some limitations, such as low search accuracy and poor path smoothness [11]. The bionic algorithms find the optimize path by using the behavior of biological structure [12]. Typical algorithms include genetic algorithm [13], ant colony optimization [14], etc. Compared with traditional algorithms, bionic algorithms have less computation and have

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利用深度强化学习的综合未知环境下自主移动机器人的路径规划

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Abstract**——**在现实场景中，未知环境下的自主移动机器人（AMR）的路径规划往往会遇到一些不可避免的问题，如对环境信息的严重依赖、推理时间长、抗干扰能力弱等。为了解决这些问题，本文提出了一种改进的基于深度强化学习的路径规划算法，以找出一类 AMR 的优化路径。首先，将AMR的路径规划描述为马尔可夫决策过程框架，并利用双深度Q网络（DDQN）来获得AMR路径规划的最优自适应解。其次，设计了与启发式功能相结合的综合奖励功能，以将 AMR 引导到目标区域。随后，设计了具有自适应ε-贪婪动作选择策略的优化深度神经网络来处理探索和利用之间的权衡，从而进一步提高AMR路径规划的全局搜索能力和收敛性能。此外，利用贝塞尔曲线理论来平滑规划路径。最后，进行比较模拟以验证我们提出的路径规划算法。结果表明，与DQN、A*、RRT和APF算法相比，我们改进的DDQN算法可以在全面的未知环境中产生更安全、更短的全局路径。同时，IDDQN算法对未知环境中的随机扰动具有很强的适应性。

Index Terms**——**自主移动机器人（AMR）、深度强化学习（DRL）、双深度Q网络（DDQN）、路径规划、路径平滑。

一、简介

近年来，自主移动机器人（AMR）以其体积小、部署灵活、独立完成任务等优点，在生物医学、太空探索、工程设备检修与维护等领域获得了广泛的应用前景[1]。路径规划是AMR领域的核心功能之一。因此，提高AMR的自主路径规划能力对于研究人员来说显得尤为重要。路径规划系统需要生成一条可行路径，同时满足多重约束：如与障碍物保持安全距离、到目标的最短距离、路径平滑度和环境干扰信息[2]、[3]。在未知环境中执行任务时，AMR 应根据从周围环境收集的信息计算并修改其可行的无碰撞路径 [4]。此外，AMR需要在存在外部环境因素潜在干扰的情况下生成符合约束的路径导航信息。因此，具有高度自主性和学习能力的路径规划方法对于AMR来说至关重要。

机器人系统的路径规划算法已经得到了大量的研究成果，大致可分为三类：1）传统算法；2）仿生算法；3）基于学习的算法，如图1所示。传统算法包括基于图、基于采样和基于数据的方法。仿生算法包括遗传算法、人工群体算法、模糊逻辑算法等。基于学习的算法包括深度强化学习（DRL）算法和元学习等。具体来说，常用的传统方法包括A星算法（A*）[5]、人工势场算法（APF）[6]和快速探索随机树算法（RRT）[7]。A*算法虽然可以有效地确定最短路径，但其计算量会随着空间的扩展呈指数增长，因此不适合复杂的连续障碍物环境[8]。APF算法采用势函数下降法来识别理想路径[9]。林等人。[10]设计了一种两级路径规划方法，采用改进的APF算法和动态窗口算法来引导机器人避开障碍物，合理规划更短、更平滑的路径。然而，随着障碍物数量的增加，APF算法会遇到环境中越来越多的零势能点，很容易陷入局部极小点而无法规划出完整的路径。RRT算法是一种基于随机采样的路径规划算法，虽然RRT算法具有较强的搜索能力和较快的搜索速度，但也存在搜索精度低、路径平滑性差等局限性[11]。仿生算法利用生物结构的行为来寻找优化路径[12]。典型的算法包括遗传算法[13]、蚁群优化[14]等。与传统算法相比，仿生算法计算量少，具有

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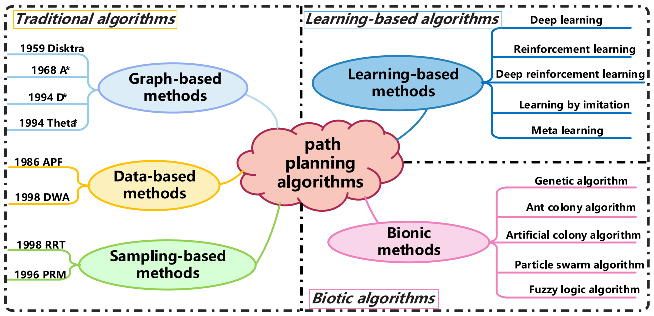

Fig. 1. Classification diagram of path-planning algorithm.

obvious advantages in dealing with complex environments. However, they still have some deficiencies, such as strong dependence on the environment and being easy to fall into local optimum points.

Accordingly, traditional methods and bionic algorithms are difficult to solve the path-planning problem without global prior environmental information, and lack adaptability to uncertainties such as system disturbances. In order to implement path planning in unknown environment, some researchers are beginning to develop smarter path-planning methods that allow AMRs to learn on their own in practice. Machine learning (ML) is one of the fastest developing AI algorithms, which aims to automatically improve the system strategy by learning from historical empirical data [15]. Liu et al. [16] designed a tiny ML algorithm to assist Internet of Unmanned Aerial Vehicles in making real-time decisions during flight, ultimately producing an efficient and secure flight path. Therefore, the application of ML methods in the field of robot navigation can make the robot learn environmental information and complete the path-planning missions autonomously.

Reinforcement learning (RL) as a branch of ML can learn and adjust the learning policy by interacting with the external environment and getting feedback information from the external environment, and RL shows great potential in overcoming dependence on the environment information [17], [18]. In [19], the Q-Learning algorithm was used to handle with the obstacle avoidance and path-planning problems of autonomous underwater vehicles in unknown environments. In view of the better path-planning ability of RL algorithm in unknown environment, some studies have successfully applied the RL in mobile robot systems. For instance, [20] proposed an algorithm combining Q-Learning and FPA, and applying it to path planning of AMR, which solved the problem of slow convergence speed of traditional Q-Learning algorithm. In [21], it incorporated heuristic search strategies and simulated annealing mechanism into Q-Learning based on Dyna architecture, which enhanced the global search performance and learning efficiency of mobile robot path planning. Nonetheless, the Q-learning algorithm stores all the state values in the environment by establishing a Q-value table and accesses this information through queries. As the dimensions of the environment and actions increase, the Q-Learning algorithm requires enormous computing resources, which reduces the learning efficiency and potentially impeding algorithm convergence [22].

DRL integrates the decision-making capability of RL and perception capability of deep learning (DL), which can bridge the gap between high-dimensional inputs and actions. In recent years, DRL has been extensively utilized in the control of unmanned aerial vehicles, autonomous underwater vehicles, and unmanned surface vehicles path planning [23], [24], [25], [26], as well as autonomous driving [27]. Additionally, several DRL-based AMR path-planning algorithms have been proposed by academics; these algorithms are essentially based on deep Q network (DQN) and its associated algorithms. Specifically, Wen et al. [28] proposed a novel path-planning method based on neural network RL path system, which enabled mobile robots to navigate to the terminal area without collision with any obstacles or robots, while this method has already succeed applied in the mobile robot experiment platform. Wu et al. [29] proposed a modified double DQN (DDQN) algorithm for autonomous steering of mobile robots, and validated its performance in various types of real-world obstacle environments. Wang et al. [30] proposed a novel free gait multicontact path-planning method based on HFG-DRL, the trained path-planning strategy can make the robot adjust its gait pattern in unstructured environment automatically, and reach the target area quickly and smoothly. Tao and Hafid [31] proposed a DeepSensing framework based on the DDQN-PER algorithm to address the depth-sensing task allocation problem, thus planed more efficient travel paths for mobile users and achieved specific targets. Jiang et al. [32] proposed an improved DQN algorithm to address the path-planning issue of hopping rover on the asteroid surface with complicated topography, and demonstrated that the path-planning algorithm has certain adaptability under randomly changing terrain. However, the direct application of the DQN algorithm to path planning gives rise to several inherent issues: 1) as the complexity of the environment model increases, the learning efficiency of DQN will become worse and the convergence speed will decrease; 2) the paths planned by the traditional DQN algorithm often exhibit inferior in safety, the AMR has the risk of colliding with the edges of obstacles; and 3) the paths trajectory planned by the traditional DQN algorithm often exhibit redundancy and lack smoothness.

Inspired by the preceding discussion, this study proposes an improved double DQN (IDDQN)-based algorithm for AMR to perform a path-planning mission in comprehensive unknown environments with random obstacles. The main contributions of this article are summarized as follows.

1) A new type of comprehensive reward function is designed to make AMR approaching the target area rapidly and safely in comprehensive unknown environments.
2) An adaptive ε-greedy action policy is designed to solve the tradeoff between exploration and exploitation. This policy can improve the learning efficiency and convergence rate of AMR path planning by combining the optimized deep neural network (DNN).
3) Multiple simulation experimental results show that the proposed IDDQN algorithm exhibits better performance than traditional methods in addressing the AMR path-planning problem. The robustness of the IDDQN

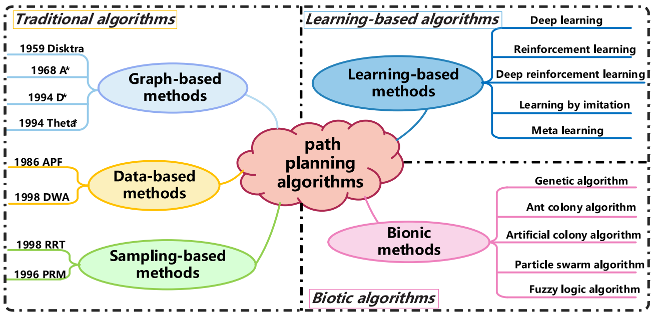

图 1 路径规划算法分类图。

应对复杂环境优势明显。但它们仍然存在一些不足，如对环境的依赖性强、容易陷入局部最优等。

因此，传统方法和仿生算法在没有全局先验环境信息的情况下很难解决路径规划问题，并且缺乏对系统扰动等不确定性的适应性。为了在未知环境中实现路径规划，一些研究人员开始开发更智能的路径规划方法，使AMR能够在实践中自行学习。机器学习（ML）是发展最快的人工智能算法之一，旨在通过学习历史经验数据来自动改进系统策略[15]。刘等人。 [16]设计了一种微型机器学习算法来协助无人机互联网在飞行过程中做出实时决策，最终产生高效且安全的飞行路径。因此，将机器学习方法应用于机器人导航领域可以使机器人学习环境信息并自主完成路径规划任务。

强化学习（RL）作为机器学习的一个分支，可以通过与外部环境交互并从外部环境获取反馈信息来学习和调整学习策略，并且RL在克服对环境信息的依赖方面显示出巨大的潜力[17]，[18]。在[19]中，Q-学习算法被用来处理未知环境下自主水下航行器的避障和路径规划问题。鉴于强化学习算法在未知环境下具有更好的路径规划能力，一些研究已成功地将强化学习应用于移动机器人系统中。例如，[20]提出了一种将Q-Learning与FPA相结合的算法，并将其应用于AMR的路径规划，解决了传统Q-Learning算法收敛速度慢的问题。在[21]中，它将启发式搜索策略和模拟退火机制纳入基于Dyna架构的Q-Learning中，增强了移动机器人路径规划的全局搜索性能和学习效率。尽管如此，Q-学习算法通过建立Q-值表来存储环境中的所有状态值，并通过查询访问这些信息。随着环境和动作维度的增加，Q-学习算法需要大量的计算资源，这降低了学习效率并可能阻碍算法收敛[22]。

DRL集成了RL的决策能力和深度学习（DL）的感知能力，可以弥合高维输入和行动之间的差距。近年来，DRL已广泛应用于无人机、自主水下航行器和无人水面航行器路径规划的控制[23]、[24]、[25]、[26]以及自动驾驶[27]。此外，学术界还提出了几种基于DRL的AMR路径规划算法；这些算法本质上基于深度 Q 网络 (DQN) 及其相关算法。具体来说，Wen 等人。 [28]提出了一种基于神经网络强化学习路径系统的新型路径规划方法，使移动机器人能够在不与任何障碍物或机器人碰撞的情况下导航到终端区域，并且该方法已经成功应用于移动机器人实验平台。吴等人。 [29]提出了一种用于移动机器人自主转向的改进的双DQN（DDQN）算法，并验证了其在各种类型的现实障碍环境中的性能。王等人。 [30]提出了一种基于HFG-DRL的新型自由步态多接触路径规划方法，训练后的路径规划策略可以使机器人在非结构化环境中自动调整步态模式，快速平稳地到达目标区域。 Tai和Hafid[31]提出了一种基于DDQN-PER算法的深度感知框架来解决深度感知任务分配问题，从而为移动用户规划更有效的行进路径并实现特定目标。江等人。文献[32]提出了一种改进的DQN算法来解决地形复杂的小行星表面跳跃漫游器的路径规划问题，并证明该路径规划算法在随机变化的地形下具有一定的适应性。然而，直接将DQN算法应用于路径规划会带来几个固有的问题：1）随着环境模型复杂度的增加，DQN的学习效率会变差，收敛速度会降低； 2）传统DQN算法规划的路径往往安全性较差，AMR存在碰撞障碍物边缘的风险； 3）传统DQN算法规划的路径轨迹往往存在冗余且缺乏平滑性。

受前面讨论的启发，本研究提出了一种改进的基于双DQN（IDDQN）的AMR算法，用于在具有随机障碍物的全面未知环境中执行路径规划任务。本文的主要贡献总结如下。

1）设计了一种新型的综合奖励函数，使AMR在综合未知环境中快速、安全地逼近目标区域。 2）自适应ε-贪婪行动策略旨在解决探索和利用之间的权衡。该策略可以通过结合优化的深度神经网络（DNN）来提高AMR路径规划的学习效率和收敛率。 3）多次仿真实验结果表明，所提出的IDDQN算法在解决AMR路径规划问题上比传统方法表现出更好的性能。 IDDQN 的稳健性

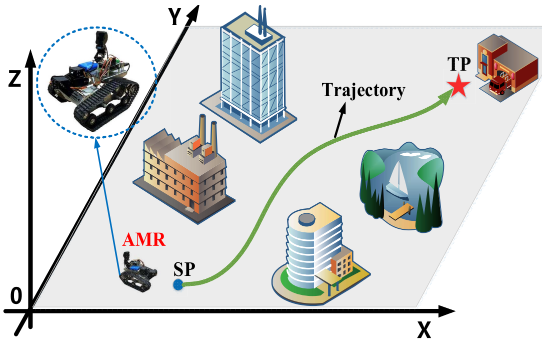

Fig. 2. Schematic of AMR path planning.

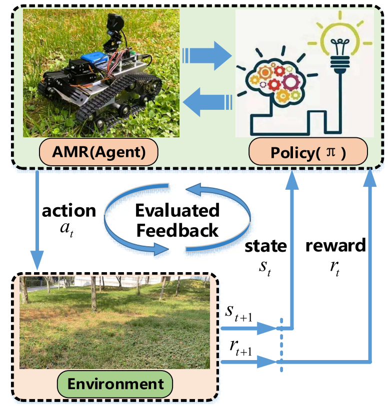

Fig. 3. AMR–environment interaction in the MDP framework.

algorithm is also verified in unknown environments with random disturbances.

The remainder of this article is arranged as follows. Section II describes the path-planning problem of AMR and introduces Markov decision process (MDP) and DDQN algorithm. Section III presents the MDP model of AMR path planning and the AMR path-planning algorithm model based on IDDQN. Section IV presents the simulation comparative experimental analysis of different path-planning algorithms, and verifies the robustness of the IDDQN algorithm in disturbed environments. The conclusion and future works are presented in Section V.

II. Basic Theory for AMR Path Planning

AMR path planning, as shown in Fig. 2, refers to a complete process that AMR can automatically generate a smooth and collision-free optimal path (green trajectory line) from start point SP) to target point (TP) considering all external environment information and multiple constraints [33].

According to the related literatures [34], [35], the AMR's path planning can be regarded as a learning process in the MDP framework. In which, the AMR takes actions to interact with the external environment and changes its state to obtain a reward; meanwhile, the AMR aims to learn the action policy that obtains the maximum cumulative reward, and finally generates the required path trajectory. Based on the aforementioned descriptions, the knowledge of MDP theory and DDQN algorithm should be first introduced.

A. Markov Decision Processes

MDP can be expressed as a five-tuple $M = [S, A, P, R]$, where S represents the finite set of states that exists in the environment, s_t represents the state at time t, A indicates the action executed by the agent (AMR), and a_t indicates the action performed at time t, P is the transfer probability, and R is the reward function. Based on the observed environmental state s_t, the AMR randomly selects and executes action a_t. Subsequently, the environment provides the reward r_{t+1} for the AMR, while updating the state from s_t to next state s_{t+1}. The AMR–environment interaction in the MDP framework is shown in Fig. 3.

Specifically, at time t, when the AMR performs action $s \in S$ in state $a \in A$, s' can be given by

$$P(s, a, s') = P(s' s, a) = \text{prob}\left[s_{t+1} = s' S_t = s, A_t = a\right]. \quad (1)$$

Throughout the learning process, AMR aims to learn the optimal policy corresponding to the maximized long-term cumulative rewards. Then, AMR can get the maximum long-term cumulative reward R_t during exploration. R_t can be given by

$$R_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \cdots = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \quad (2)$$

where r_t represents the AMR's reward at time t. γ is the discount rate ($\gamma \in [0, 1]$), and γ determines the future return value.

To obtain the long-term relationship between the current action and the future reward, the action-value function $Q_\pi(s, a)$ is formulated using the total potential reward under the optimal policy π, which can be expressed by

$$Q_\pi(s, a) = E_\pi[R_t S_t = s, A_t = a] = E_\pi\left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} | s_t, a_t\right] \quad (3)$$

where E_π represents the mathematical expectation of the random variable under the probability distribution π.

B. Double Deep Q Network

In this section, the DQN algorithm is utilized to solve the AMR path-planning decision problems. DQN, as a prominent algorithm in DRL, effectively combines Q-Learning and neural network technique to acquire an optimal control strategy from a high-dimensional and complex state space, without depending on prior information of the environment [36]. Consequently, the utilization of DQN enables AMR to produce an optimal path in complex obstacle-laden environments with incomplete external environmental information.

The DQN algorithm employs neural networks to approximate the Q-value function, as depicted in (3). The overall learning architecture of the DQN algorithm is illustrated in Fig. 4. Notably, the incorporation of target networks and

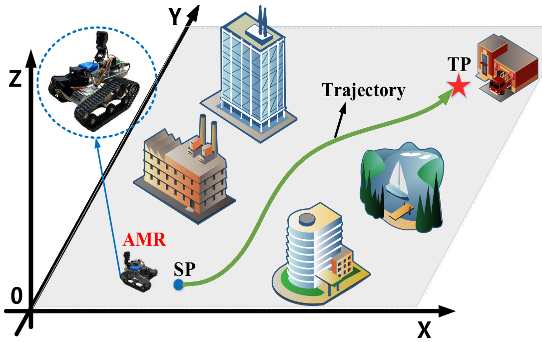

图2.AMR路径规划示意图。

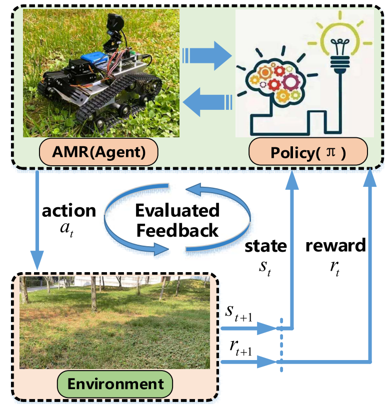

图 3. MDP 框架中的 AMR-环境交互。

算法还在具有随机干扰的未知环境中得到验证。

本文的其余部分安排如下。第二节描述了AMR的路径规划问题，并介绍了马尔可夫决策过程（MDP）和DDQN算法。第三节介绍了AMR路径规划的MDP模型和基于IDDQN的AMR路径规划算法模型。第四节介绍了不同路径规划算法的仿真对比实验分析，并验证了IDDQN算法在干扰环境中的鲁棒性。结论和未来的工作在第五节中介绍。

二. AMR 路径规划的基本理论

AMR路径规划如图2所示，是指AMR考虑所有外部环境信息和多重约束，自动生成从起点SP）到目标点（TP）的平滑且无碰撞的最优路径（绿色轨迹线）的完整过程[33]。

根据相关文献[34]、[35]，AMR的路径规划可以看作是MDP框架中的学习过程。其中，AMR采取行动与外部环境交互，改变自身状态以获得奖励；同时，AMR的目标是学习获得最大累积奖励的行动策略，最终生成所需的路径轨迹。基于上述描述，首先介绍MDP理论和DDQN算法的知识。

A. *Markov Decision Processes*

MDP可以表示为五元组$M = [S, A, P, R]$，其中S表示环境中存在的有限状态集，s_t表示时间t的状态，A表示代理执行的动作（AMR），a_t表示时间t执行的动作，P是转移概率，R是奖励函数。根据观察到的环境状态s_t，AMR随机选择并执行动作a_t。随后，环境为 AMR 提供奖励 r_{t+1}，同时将状态从 s_t 更新到下一个状态 s_{t+1}。 MDP框架中的AMR-环境交互如图3所示。

具体来说，在时间 t，当 AMR 在状态 $a \in A$ 下执行操作 $s \in S$ 时，s' 可以由下式给出

$$P(s, a, s') = P(s's, a) = \text{prob}[s_{t+1} = s'S_t = s, A_t = a]. \quad (1)$$

在整个学习过程中，AMR旨在学习与最大化长期累积奖励相对应的最优策略。然后，AMR可以在探索过程中获得最大的长期累积奖励R_t。 R_t 可以由下式给出

$$R_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \cdots = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \quad (2)$$

其中r_t表示AMR在时间t的奖励。 γ 是折扣率 ($\gamma \in [0, 1]$)，γ 决定未来的回报值。

为了获得当前行动和未来奖励之间的长期关系，使用最优策略π下的总潜在奖励来制定行动价值函数$Q_\pi(s, a)$，其可以表示为

$$Q_\pi(s, a) = E_\pi[R_t S_t = s, A_t = a] = E_\pi\left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} | s_t, a_t\right] \quad (3)$$

其中E_π表示随机变量在概率分布π下的数学期望。

B. *Double Deep Q Network*

在本节中，DQN算法用于解决AMR路径规划决策问题。 DQN作为DRL中的一种重要算法，有效地结合了Q学习和神经网络技术，从高维复杂的状态空间中获取最优控制策略，而不依赖于环境的先验信息[36]。因此，利用 DQN 使 AMR 能够在外部环境信息不完整的复杂障碍物环境中生成最佳路径。

DQN 算法采用神经网络来逼近 Q 值函数，如（3）所示。 DQN 算法的整体学习架构如图 4 所示。值得注意的是，目标网络和

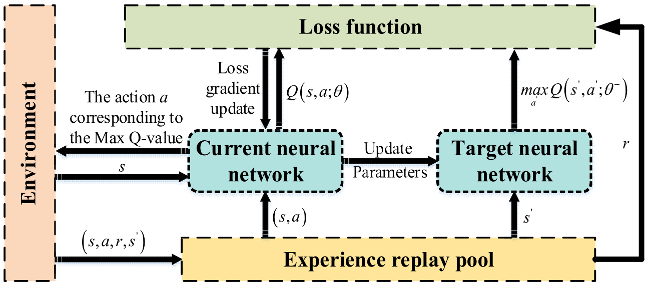

Fig. 4. Overall DQN architecture.

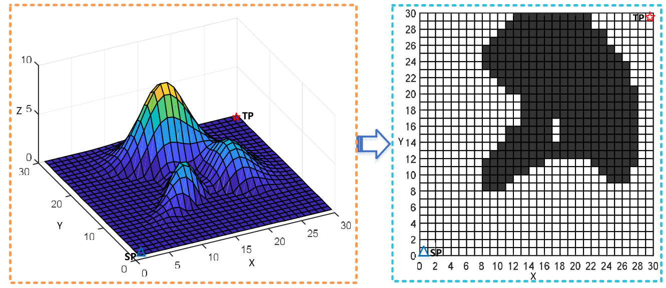

Fig. 5. Example of the environment model (left: 3-D environment and right: 2-D gride environment).

empirical replay mechanisms significantly enhances training accuracy and stability [37]. During the training process, the current network generates experience data (current state s, action a, reward r, and next state s') through interactive learning with the environment, and stores it in the experience replay pool with a certain capacity (if the storage is full, the original experience is overwritten sequentially starting from the first group). According to the empirical data obtained by the current network, the parameters of the target network are constantly updated and optimized, and finally the action strategy with the maximum reward is obtained.

The difference between the current network and the target network is how to select the input vector and the network update period. To be specific, the current network takes the current state s as the input and updates it at each iteration; the target network takes the next state s' as input and periodically copies the parameters of the current network to the target network. In each training step, the time-series difference method is used to compute the corresponding target Q-value ($Q_{\text{Target}}^{\text{DQN}}$) of the target network, which can be expressed by

$$Q_{\text{Target}}^{\text{DQN}} = r + \gamma \max_{a'} Q\big(s', a'; \theta^{-}\big) \tag{4}$$

where $\max_{a'} Q(s', a'; \theta^{-})$ is the maximum Q-value output by the target network, $a'$is the action taken by the AMR in state s', and θ^{-} denotes the weight parameter of the target network.

The mean square deviation between the Q-value predicted by the current network and the Q-value predicted by the target network is defined as the loss function $L(\theta)$, which is formed as

$$L(\theta) = E\left[\left(Q_{\text{Target}}^{\text{DQN}} - Q(s, a; \theta)\right)^{2}\right] \tag{5}$$

where $Q(s, a; \theta)$ denotes the output of the current network and θ denotes the weight parameter of the current network.

By decoupling the calculation of target Q-value ($Q_{\text{Target}}^{\text{DQN}}$) from the selection of actions in the DQN algorithm, DDQN algorithm can further eliminate the possible overestimation problem and generate a more reliable and stable learning process [38].

Instead of directly obtaining the maximum Q-value from the target network, DDQN is first employed to find the maximum Q-value based on the current network. Then, the target Q-value ($Q_{\text{Target}}^{\text{DDQN}}$) corresponding to the action is calculated using the target network with parameter θ^{-}, as shown in (6), and the selected action is evaluated as

$$Q_{\text{Target}}^{\text{DDQN}} = r + \gamma Q'\left(s', \underset{a'}{\text{argmax}} Q\big(s', a; \theta\big); \theta^{-}\right) \tag{6}$$

where $\text{argmax}_{a'} Q(s', a; \theta)$ is the action corresponding to the maximum output Q-value of the target network, the predicted value of the current network is $Q'(s', \text{argmax}_{a'} Q(s', a; \theta); \theta^{-})$.

III. Path-Planning Synthesis

In this section, the AMR path-planning problem is first reformulated as an MDP model, where the MDP mainly contains state space, action space, and reward function. Then, an improved DDQN algorithm is proposed, to solve the MDP model of AMR path planning, and the optimal path is obtained by the path smoothing method.

A. MDP of AMR Path Planning

1) State Space: The grid-based metric model is mainly used to establish the environment map [39], [40]. Herein, the feature information of the 3-D environment is mapped into a 2-D grid of equal size, as illustrated in Fig. 5, enabling AMR to perform path planning in a 30 m*30 m 2-D grid environment. According to the distribution of obstacles in the environment, the obstacles are represented by black grids, the free movement area (feasible state of the agent) of the AMR are represented by white grid, and the width of each grid is 1 m. To facilitate the construction of the simulation environment map, obstacles that do not completely occupy a grid will be considered as a complete obstacle grid.

Since the Euclidean distance $\text{TD}_i (i = 1, 2, \ldots, t)$ between AMR and the target should be considered when calculating the path length, hence the position coordinate (x, y) of AMR and TD_i are introduced to the state space. Moreover, because obstacle avoidance is crucial for path planning and assessing movement safety, the Euclidean distance between the AMR and the obstacles identified by the sensor are defined as OD_i $(i = 1, 2, \ldots, t)$, as shown in Fig. 6. Both TD_i and OD_i represent the current environmental information obtainable within a certain distance around the AMR at time i. The state

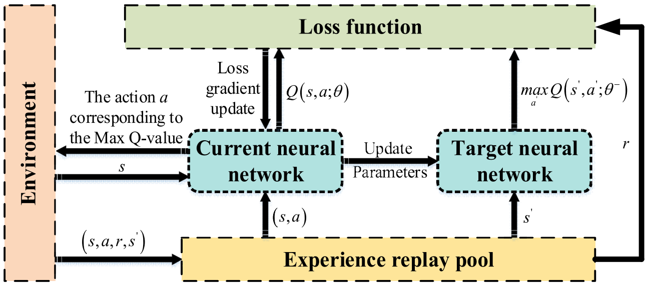

图 4. 总体 DQN 架构。

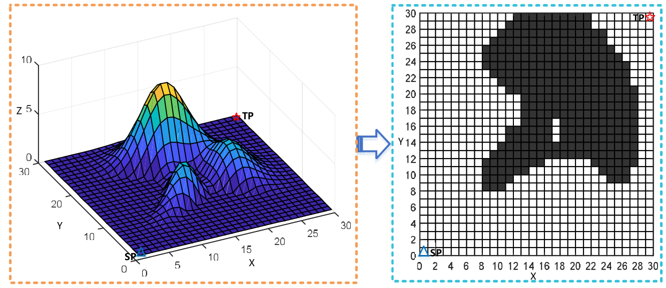

图 5. 环境模型示例（左：3-D 环境，右：2-D 网格环境）。

经验重播机制显着提高了训练的准确性和稳定性[37]。在训练过程中，当前网络通过与环境交互学习，生成经验数据（当前状态s、动作a、奖励r、下一个状态s'），并存储在一定容量的经验回放池中（如果存储满了，则从第一组开始依次覆盖原有经验）。根据当前网络获得的经验数据，不断更新和优化目标网络的参数，最终得到最大的动作策略。获得奖励。

当前网络和目标网络的区别在于如何选择输入向量和网络更新周期。具体来说，当前网络将当前状态s作为输入，并在每次迭代时更新它；目标网络将下一个状态s'作为输入，并定期将当前网络的参数复制到目标网络。在每个训练步骤中，采用时间序列差分法计算目标网络对应的目标Q-值($Q_{\text{Target}}^{\text{DQN}}$)，可以表示为

$$Q_{\text{Target}}^{\text{DQN}} = r + \gamma \max_{a'} Q\left(s', a'; \theta^-\right) \tag{4}$$

其中$\max_{a'} Q(s', a'; \theta^-)$是目标网络输出的最大$Q$值，$a'$是AMR在状态$s'$下采取的动作，$\theta^-$表示目标网络的权重参数。

当前网络预测的Q值与目标网络预测的Q值之间的均方偏差定义为损失函数$L(\theta)$，其形式为

$$L(\theta) = E\left[\left(Q_{\text{Target}}^{\text{DQN}} - Q(s, a; \theta)\right)^2\right] \tag{5}$$

其中$Q(s, a; \theta)$表示当前网络的输出，θ表示当前网络的权重参数。

通过将目标Q-值($Q_{\text{Target}}^{\text{DQN}}$)的计算与DQN算法中动作的选择解耦，DDQN算法可以进一步消除可能的高估问题并生成更可靠和稳定的学习过程[38]。

DDQN 不是直接从目标网络获取最大 Q 值，而是首先根据当前网络查找最大 Q 值。然后，使用以下公式计算对应于该动作的目标Q值($Q_{\text{Target}}^{\text{DDQN}}$)具有参数$\theta^-$的目标网络，如(6)所示，并且所选动作被评估为

$$Q_{\text{Target}}^{\text{DDQN}} = r + \gamma Q'\left(s', \operatorname*{argmax}_{a'} Q\left(s', a; \theta\right); \theta^-\right) \tag{6}$$

其中$\operatorname{argmax}_{a'} Q(s', a; \theta)$是目标网络的最大输出$Q$值对应的动作，当前网络的预测值为$Q'(s', \operatorname{argmax}_{a'} Q(s', a; \theta); \theta^-)$。

三. 路径规划综合

在本节中，AMR路径规划问题首先被重新表述为MDP模型，其中MDP主要包含状态空间、动作空间和奖励函数。然后，提出一种改进的DDQN算法，求解AMR路径规划的MDP模型，并通过路径平滑方法获得最优路径。

A. MDP of AMR Path Planning

1) State Space: 基于网格的度量模型主要用于建立环境地图[39]，[40]。这里，将3D环境的特征信息映射到等尺寸的2D网格中，如图5所示，使得AMR能够在30m*30m的2D网格环境中进行路径规划。根据环境中障碍物的分布情况，障碍物用黑色网格表示，AMR的自由运动区域（智能体的可行状态）用白色网格表示，每个网格的宽度为1 m。为了便于构建模拟环境地图，未完全占据网格的障碍物将被视为完整的障碍物网格。

由于计算路径长度时要考虑AMR与目标之间的欧氏距离$\text{TD}_i (i = 1, 2, \ldots, t)$，因此将AMR和$\text{TD}_i$的位置坐标($x$，$y$)引入状态空间。此外，由于避障对于路径规划和评估运动安全至关重要，因此AMR与传感器识别的障碍物之间的欧氏距离定义为OD_i ($i = 1, 2, \ldots, t$)，如图6所示。TD_i和OD_i都表示在时间i时AMR周围一定距离内可获得的当前环境信息。国家

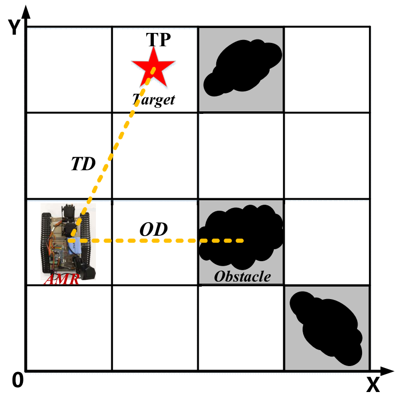

Fig. 6. State space.

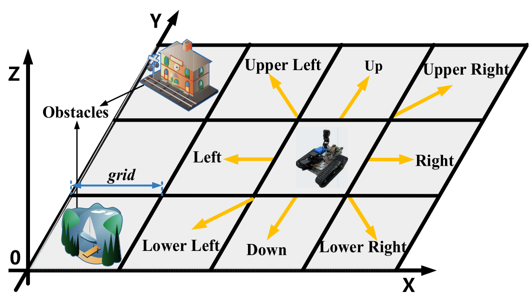

Fig. 7. AMR action space.

space equation of the AMR can be expressed as

$$S = \begin{bmatrix} x_1 & y_1 & \mathrm{TD}_1 & \mathrm{OD}_1 \\ x_2 & y_2 & \mathrm{TD}_2 & \mathrm{OD}_2 \\ \vdots & \vdots & \vdots & \vdots \\ x_t & y_t & \mathrm{TD}_t & \mathrm{OD}_t \end{bmatrix}_{t\times 4}. \quad (7)$$

2) Action Space: Generally, the motion state of AMR contains starting, stopping, linear motion, and steering, which are controlled by servo motors, and the velocity of AMR is presumed to remain constant during path planning. To simplify the motion model of the AMR, the action space of the AMR can be described as eight actions, i.e., $\mathbf{A} = \{$up, down, left, right, upper right, lower right, upper left and lower left$\}$, the specific action space is shown in Fig. 7.

Based on this, the AMR's transfer rules in the grid environment can be expressed as

$$\begin{cases} (x', y') = (x + \mathrm{grid}, y) \\ (x', y') = (x, y + \mathrm{grid}) \\ (x', y') = (x - \mathrm{grid}, y) \\ (x', y') = (x, y - \mathrm{grid}) \\ (x', y') = (x + \mathrm{grid}, y + \mathrm{grid}) \\ (x', y') = (x + \mathrm{grid}, y - \mathrm{grid}) \\ (x', y') = (x - \mathrm{grid}, y + \mathrm{grid}) \\ (x', y') = (x - \mathrm{grid}, y - \mathrm{grid}) \end{cases} \quad (8)$$

where (x, y) denotes the current state of the AMR, (x', y') denotes the next state of the AMR, and *grid* represents the wide of one unit grid. It is worth emphasizing that each action performed by the AMR in the grid environment can only move it to neighboring states.

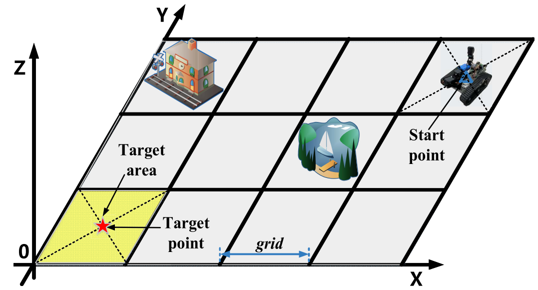

Fig. 8. Target area diagram.

3) Design of Reward Functions: In the MDP framework, the reward function is used to evaluate the value of the agent's (AMR) action a. In most previous studies, when approaching to the target area, AMR will receive a positive reward, when colliding with obstacles, AMR will receive a negative reward. However, the reward values for other states are set as zero, which led to the problem of sparse rewards in RL algorithms. Therefore, we introduce a composite reward function R_t to make the rewards received by the agent changing as the state transitions during the algorithm training process. The composite reward function can be represented as

$$R_t = r_{\tau 1} + r_{\tau 2} + r_{\tau 3} + r_{\tau 4} \quad (9)$$

where $r_{\tau 1}$ is the target reward function, $r_{\tau 2}$ is the distance reward function, $r_{\tau 3}$ is the boundary reward function, and $r_{\tau 4}$ is the obstacle reward function.

In a detailed description, $r_{\tau 1}$ is usually set as a positive value and is used to motivate AMR to approach the TP, which is calculated as

$$r_{\tau 1} = \lambda_1 \left(\mathrm{TD} \leq \frac{\sqrt{2}}{2} \mathrm{grid} \right) \quad (10)$$

where λ_1 represents the positive reward value, which can be obtained when AMR reaches the target area. To improve the MDP model training efficiency, the mission can be regarded as completed when the AMR reaches the yellow area around the TP, as shown in Fig. 8.

The distance reward function $r_{\tau 2}$ takes the Euclidian distance as the heuristic function term [41], which can reduce the blindness of the DDQN algorithm in the environment search process and enhance the planning efficiency. The formula is given by

$$r_{\tau 2} = \lambda_2 \left(\sqrt{\left(x_k - x_{\mathrm{target}}\right)^2 + \left(y_k - y_{\mathrm{target}}\right)^2} \right) \quad (11)$$

where λ_2 is a negative constant that is used to control the magnitude of the distance reward function, (x_k, y_k) denotes the state of the AMR at time k, and $(x_{\mathrm{target}}, y_{\mathrm{target}})$ represents the target state. This implies that if the AMR is closer to the target area, it will receive a smaller negative reward value. Consequently, $r_{\tau 2}$ can facilitate the AMR's rapid arrival in the target area.

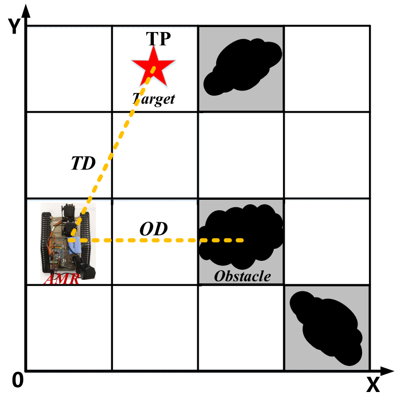

图 6. 状态空间。

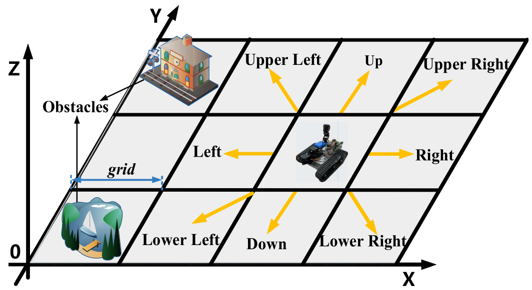

图 7.AMR 操作空间。

AMR的空间方程可表示为

$$S=\begin{bmatrix} x_1 & y_1 & \mathrm{TD}_1 & \mathrm{OD}_1 \\ x_2 & y_2 & \mathrm{TD}_2 & \mathrm{OD}_2 \\ \vdots & \vdots & \vdots & \vdots \\ x_t & y_t & \mathrm{TD}_t & \mathrm{OD}_t \end{bmatrix}_{t\times 4}. \tag{7}$$

2) Action Space: 通常，AMR的运动状态包括启动、停止、直线运动和转向，这些状态由伺服电机控制，并且在路径规划过程中假设AMR的速度保持恒定。为了简化AMR的运动模型，AMR的动作空间可以描述为8个动作，即A = {上、下、左、右、右上、右下、左上、左下}，具体动作空间如图7所示。

基于此，AMR在网格环境下的传输规则可以表示为

$$\begin{cases} (x',y') = (x+\mathrm{grid},y) \\ (x',y') = (x,y+\mathrm{grid}) \\ (x',y') = (x-\mathrm{grid},y) \\ (x',y') = (x,y-\mathrm{grid}) \\ (x',y') = (x+\mathrm{grid},y+\mathrm{grid}) \\ (x',y') = (x+\mathrm{grid},y-\mathrm{grid}) \\ (x',y') = (x-\mathrm{grid},y+\mathrm{grid}) \\ (x',y') = (x-\mathrm{grid},y-\mathrm{grid}) \end{cases} \tag{8}$$

其中（x，y）表示AMR的当前状态，(x',y')表示AMR的下一个状态，*grid*表示一个单位网格的宽度。值得强调的是，AMR在网格环境中执行的每个动作只能将其移动到邻近的状态。

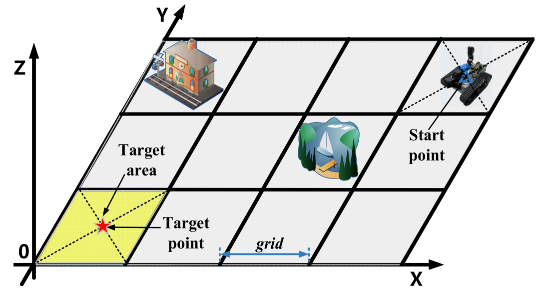

图 8. 目标区域图。

3) Design of Reward Functions: 在MDP框架中，奖励函数用于评估代理（AMR）动作a的价值。在之前的大多数研究中，当接近目标区域时，AMR会获得正奖励，当碰撞障碍物时，AMR会获得负奖励。然而，其他状态的奖励值设置为零，这导致了强化学习算法奖励稀疏的问题。因此，我们引入了复合奖励函数R_t，使智能体在算法训练过程中收到的奖励随着状态转换而变化。复合奖励函数可以表示为

$$R_t = r_{\tau 1} + r_{\tau 2} + r_{\tau 3} + r_{\tau 4} \tag{9}$$

其中$r_{\tau 1}$是目标奖励函数，$r_{\tau 2}$是距离奖励函数，$r_{\tau 3}$是边界奖励函数，$r_{\tau 4}$是障碍物奖励函数。

详细描述中，$r_{\tau 1}$通常设置为正值，用于激励AMR逼近TP，计算公式为

$$r_{\tau 1} = \lambda_1\left(\mathrm{TD} \leq \frac{\sqrt{2}}{2}\mathrm{grid}\right) \tag{10}$$

其中λ_1表示正奖励值，当AMR到达目标区域时可以获得。为了提高MDP模型训练效率，当AMR到达TP周围的黄色区域时，可以认为任务完成，如图8所示。

距离奖励函数$r_{\tau 2}$以欧几里德距离作为启发函数项[41]，可以减少DDQN算法在环境搜索过程中的盲目性，提高规划效率。公式由下式给出

$$r_{\tau 2} = \lambda_2\left(\sqrt{\left(x_k - x_{\mathrm{target}}\right)^2 + \left(y_k - y_{\mathrm{target}}\right)^2}\right) \tag{11}$$

其中λ_2是一个负常数，用于控制距离奖励函数的大小，(x_k, y_k)表示AMR在时间k的状态，$(x_{\mathrm{target}}, y_{\mathrm{target}})$表示目标状态。这意味着如果AMR距离目标区域越近，它将获得较小的负奖励值。因此，$r_{\tau 2}$可以促进AMR快速到达目标区域。

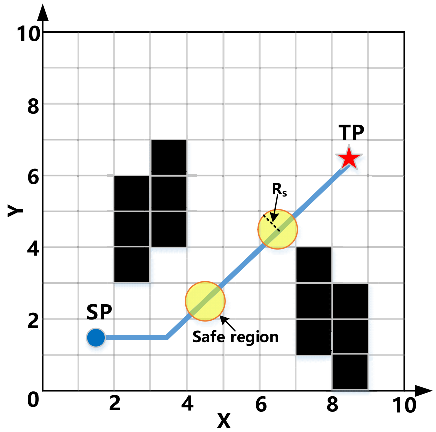

Fig. 9. Schematic of minimum safe distance.

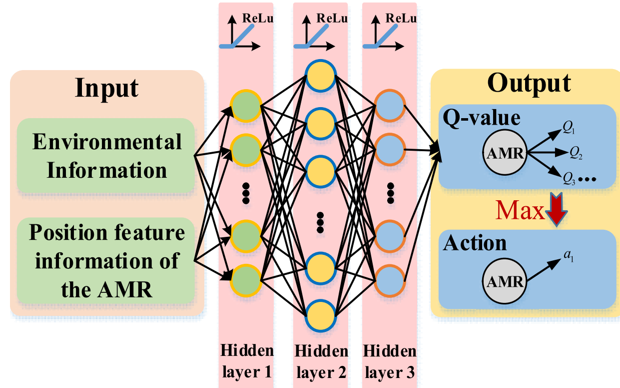

Fig. 10. Neural network structure of the current and target networks.

The boundary reward function $r_{\tau 3}$ can confine the AMR motion within the boundaries of the environment, which is given by

$$r_{\tau 3} = \begin{cases} \lambda_3, & \begin{pmatrix} x > \max(x) \text{ or } x < \min(x) \\ y > \max(y) \text{ or } y < \min(y) \end{pmatrix} \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

In (12), λ_3 represents a negative constant value, $\max(x)$ and $\min(x)$ are the maximum and minimum horizontal coordinates of the environment, and $\max(y)$ and $\min(y)$ are the maximum and minimum vertical coordinates of the environment.

Due to the nonnegligible width of the AMR's body, the obstacle reward function $r_{\tau 4}$ can effectively avoid the risk of collision between the AMR and the edges of obstacles during its actual motion process. $r_{\tau 4}$ is expressed as

$$r_{\tau 4} = \begin{cases} \lambda_4, & \text{OD} < R_s \\ 0, & \text{OD} \geq R_s \end{cases} \tag{13}$$

where λ_4 represents a negative value, the minimum safe distance OD between AMR and the obstacle is set as R_s, and the circular area with radius R_s is used as the minimum safe area, the details are shown in Fig. 9. When the distance between the AMR and obstacles is less than $R_s (R_s < 0.6$ m), $r_{\tau 4}$ gives a negative reward value λ_4 to the AMR; when the obstacle occurs outside the minimum safe area of the AMR ($R_s \geq 0.6$ m), there is no risk of collision, and the value of the reward function $r_{\tau 4}$ is 0.

B. Algorithms Proposed Path-Planning Algorithm of AMR

1) Deep Neural Network: As shown in Fig. 10, the DDQN algorithm architecture adopts DNN to approximate the state–action value function [42]. Furthermore, both the current network and target network adopt the identical DNN structure. The DNN architecture consists of one input layer, three hidden layers, and one output layer, and the network parameters are learned in a supervised manner. The neural network's input data consists of two parts: 1) the environment information scanned by the sensor and 2) position feature information of the AMR. These data are processed through the fully connected layer, with the resultant computations transmitted to the output layer. The output layer providing the Q-value corresponding to the set of output actions, and ultimately output the action corresponding to the maximum Q-value. Within this network architecture, the rectified linear unit (ReLu) is employed as the activation function for both the input layer and each of the hidden layers.

The *ReLu* functions can be expressed as

$$h(\xi) = \begin{cases} \xi, & \xi > 0 \\ 0, & \xi \leq 0. \end{cases} \tag{14}$$

To make DNN maintain better learning efficiency, the learning rate is set to 0.0025 at the initiation of training process. Then, the adaptive moment estimation (Adam) algorithm is utilized to optimize the learning rate, which can achieve adaptive updating of the learning rate [43]. Meanwhile, the Adam incorporates momentum term to improve stochastic gradient descent for neural networks. Subsequently, the current network is optimized in combination with the preventing gradient explosion method, which improves the prediction ability of the DDQN algorithm.

2) Adaptive Action Selection Policy: The balance between exploration and exploitation is a challenge for RL algorithms. Agent (AMR) needs to make full use of the learned experience and constantly explore the environment to obtain a more effective long-term action policy. In prior research, a common approach to tackle this predicament has been the utilization of a fixed ε-greedy policy [44], AMR selects an action corresponding to the maximum action value function in its experience or randomly chooses an action with a certain probability. Nevertheless, an overreliance on historical experience may lead to the issue of AMR getting trapped in local optima, preventing it from attaining globally optimal solutions. Conversely, excessive exploration may diminish the training efficiency of AMR.

To make a tradeoff between exploration and exploitation, a probability-based nonlinear adaptive ε-greedy action selection policy has been formulated in this article, where the probability ε undergoes adaptive adjustments with the variation of the training episode. This policy can guide AMR to extensively explore the environment during the early stages of training and then acquire a sufficient number of training samples. In the later stages of training, the DDQN algorithm makes full use of the optimal policy to accelerate the convergence of

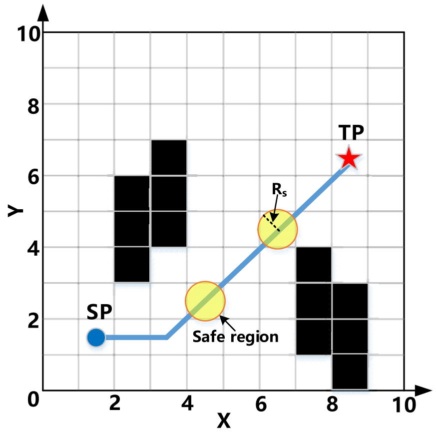

图9.最小安全距离示意图。

边界奖励函数 $r_{\tau 3}$ 可以将 AMR 运动限制在环境边界内，由下式给出

$$r_{\tau 3} = \begin{cases} \lambda_3, \begin{pmatrix} x > \max(x) \text{ or } x < \min(x) \\ y > \max(y) \text{ or } y < \min(y) \end{pmatrix} \\ 0, \quad \text{otherwise.} \end{cases} \tag{12}$$

(12)中，λ_3表示负常数值，$\max(x)$和$\min(x)$是环境的最大和最小水平坐标，$\max(y)$和$\min(y)$是环境的最大和最小垂直坐标。

由于AMR本体的宽度不可忽略，障碍物奖励函数$r_{\tau 4}$可以有效避免AMR在实际运动过程中与障碍物边缘发生碰撞的风险。 $r_{\tau 4}$ 表示为

$$r_{\tau 4} = \begin{cases} \lambda_4, \text{ OD} < R_s \\ 0, \quad \text{OD} \geq R_s \end{cases} \tag{13}$$

其中λ_4代表负值，设置AMR与障碍物之间的最小安全距离OD为R_s，以半径为R_s的圆形区域作为最小安全区域，具体如图9所示。当AMR与障碍物之间的距离小于R_s($R_s < 0.6$ m)时，$r_{\tau 4}$给予AMR负奖励值λ_4；当障碍物出现在AMR ($R_s \geq 0.6$ m)的最小安全区域之外时，不存在碰撞风险，奖励函数$r_{\tau 4}$的值为0。

B. Algorithms Proposed Path-Planning Algorithm of AMR

1) Deep Neural Network: 如图10所示，DDQN算法架构采用DNN来逼近状态-动作值函数[42]。此外，当前网络和目标网络都采用相同的DNN结构。 DNN 架构由 1 个输入层、3 个隐藏层和 1 个输出层组成，网络参数以监督方式学习。神经网络的输入数据由两部分组成：1）传感器扫描的环境信息和2）AMR的位置特征信息。这些数据通过全连接层进行处理，并将计算结果传输到输出层。输出层提供 Q 值

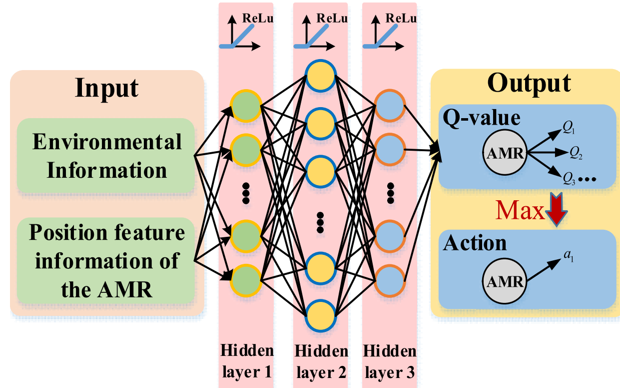

图 10. 当前网络和目标网络的神经网络结构。

对应输出动作集合，最终输出最大Q值对应的动作。在该网络架构中，整流线性单元（ReLu）被用作输入层和每个隐藏层的激活函数。

ReLu 函数可以表示为

$$h(\xi) = \begin{cases} \xi, \ \xi > 0 \\ 0, \ \xi \leq 0. \end{cases} \tag{14}$$

为了使DNN保持更好的学习效率，在训练过程开始时将学习率设置为0.0025。然后，利用自适应矩估计（Adam）算法来优化学习率，可以实现学习率的自适应更新[43]。同时，Adam 结合了动量项来改进神经网络的随机梯度下降。随后结合防止梯度爆炸的方法对当前网络进行优化，提高了DDQN算法的预测能力。

2) Adaptive Action Selection Policy: 探索和利用之间的平衡是强化学习算法面临的挑战。 Agent（AMR）需要充分利用学到的经验，不断探索环境，以获得更有效的长期行动策略。在先前的研究中，解决这种困境的常见方法是利用固定的ε-贪婪策略[44]，AMR选择与其经验中的最大动作价值函数相对应的动作，或者以一定的概率随机选择一个动作。然而，过度依赖历史经验可能会导致AMR陷入局部最优，从而无法获得全局最优解。相反，过度探索可能会降低 AMR 的训练效率。

为了在探索和利用之间进行权衡，本文提出了一种基于概率的非线性自适应 ε-贪婪动作选择策略，其中概率 ε 随着训练集的变化进行自适应调整。该策略可以引导 AMR 在训练的早期阶段广泛探索环境，然后获取足够数量的训练样本。在训练后期，DDQN算法充分利用最优策略来加速收敛

Algorithm 1 Adaptive ε-Greedy Policy

1: Generate a random probability value p, $p \in (0, 1)$
2: **If** $p < \varepsilon_k$ **then**
3: a = *random action*
4: **Else if** $p \geq \varepsilon_k$
5: $a = \underset{a}{argmax}Q(s, a)$
6: **End**

the algorithm. The adaptive action selection policy can be expressed as

$$\varepsilon_k = \varepsilon_f + \frac{\varepsilon_i - \varepsilon_f}{1 + e^{\frac{k}{\varepsilon_d}}} \tag{15}$$

where ε_k represents the kth episode greedy factors, ε_i represents initial greedy factors, $\varepsilon_\mathbf{f}$ represents final greedy factors, and ε_d represents the attenuation rate of ε-greedy. The pseudocode for the adaptive ε-greedy policy in the kth episode is given in Algorithm 1.

3) Path Smoothing: The output action of the DDQN algorithm is discrete, whereas the steering operation of the AMR in practical scenarios is continuous. To better align the planned paths more closely with the practical requirements of the AMR, it is needed to use Bezier curve to smooth the planned path, because it has such advantages as low programming difficulty, low computational cost, and strong trajectory continuity [45], [46]. Therefore, Bezier curve is introduced into the DDQN algorithm framework to realize end-to-end smooth path planning. The expressions for calculating Bezier curve from first to third order are given by

$$p_i^k = \begin{cases} p_i^0, & i = 0, 1, 2, \ldots, n-k \\ (1-t)p_i^{k-1} + tp_{i+1}^{k-1}, & i = 0, 1, 2, \ldots, n-k \end{cases} \tag{16}$$

where p_i^0 represents the ith initial control point, p_i^k represents the ith K-order control point (k=1,2…,n), and t represents the proportionality coefficient.

According to (16), the calculation formula of n-order control points always includes two n-1-order control points, so the calculation formula of n-order control points can be finally calculated from the initial control points. The second-order Bezier curve can ensure the smoothness of the path and has faster calculation speed than the higher order Bezier curve. Therefore, multiple second-order Bezier curves are concatenated to smooth the turning points of the path, and the calculation expression is formed as

$$B(t) = \left(1 - t^2\right)p_i^0 + 2(1-t)p_{i+1}^0 + t^2 p_{i+2}^0, t \in [0, 1] \tag{17}$$

where p_i^0, p_{i+1}^0, and p_{i+2}^0 represent three consecutive initialization control points.

4) AMR Path-Planning Algorithm Process: In summary, the flowchart of the IDDQN algorithm applied to AMR path planning is shown in Fig. 10. The pseudocode of the algorithm is shown in Algorithm 2, and the digits on the linking lines in Fig. 11 correspond to the flow in Algorithm 2. The flow of Algorithm 2 is depicted roughly as follows.

Step 1: First, training parameters are set, which mainly include state space, action set, initial learning rate, etc.;

Algorithm 2 IDDQN-Based AMR Path-Planning Algorithm

1: **Input**: number of interaction N; action set A; learning rate α; discount factor γ; experience replay pool maximum size D; mini-batch M; current neural network value Q; target neural network value Q'.
2: Initialize network parameters$\theta$and θ', let $\theta' = \theta$, and initialize the replay pool R.
3: Set maximum step n_{max}.
4: **For** episode=1, **do**
5: Initialize the environment space and obtain the starting state$s = (x_0, y_0)$.
6: **For** step=1, **do**
7: S is the input of the current state, get all the Q-value outputs corresponding to the current state, and use adaptive ε-greedy policy to select action corresponding to the Q-value. Set $a_t = \begin{cases} arg\ \underset{a'}{max}Q(s, a; \theta)\ with\ probability\ 1 - \varepsilon \\ random\ action \quad otherwise \end{cases}$
8: The action a is executed by AMR, and the AMR interacts with the external environment to receive the reward r and the next state s'.
9: Store (s, a, r, s') into the experience replay pool.
10: Let $s = s'$.
11: Random sample mini-batch of R transition (s_i, a_i, r_i, s_i') from D, and calculate y_i of the target Q-value.
12: Set $y_i = \begin{cases} r_i \quad if\ \ episode\ terminates\ at\ step\ i \\ r_i + \gamma Q'(S', arg\ \underset{a'}{max}Q(s', a'; \theta); \theta^-)\ otherwise \end{cases}$
13: Calculate the loss function $L(\theta) = E[(y_i - Q(s, a; \theta))^2]$, and update the parameter θ by the gradient direction reverse propagation of the neural network.
14: Replace target parameters: $\theta^- = \tau\theta + (1-\tau)\theta^-$.
15: If $s'$is the target state, the current interaction is finished, and then use Bezier curve to smooth the path. Otherwise, entry into step 7.
16: **Output**: Set of optimal path vectors.
17: **End for**
18: **End for**

second, the current neural network and the target neural network are established; and third, randomly initialize the parameters θ and θ^-, and $\theta = \theta^-$ (line 2).

Step 2: The process of AMR exploring the environment (lines 5–8). At the beginning of each episode, the environment information is initialized, and during the exploration process, the current state is inputted to the current network and the Q-value of the relevant action is output (lines 5 and 6). Action a is obtained and executed using the improved adaptive ε-greedy policy (line 7). The AMR is interacted with the environment to receive rewards r and next states s (line 8).

Step 3: The updating process of the current and target neural network (lines 9–14). First, the diversified sample data obtained during the discovery process is saved in the experience replay data pool, and then mini-batches of empirical data is randomly sampled from the experience pool to update the current network (lines 9–11). The target Q-value is updated in (line 12). Second, based on the backpropagation of the gradient of the neural network, the loss function is computed and the

Algorithm 1 Adaptive ε-Greedy Policy

1: Generate a random probability value p, $p \in (0, 1)$
2: **If** $p < \varepsilon_k$ **then**
3: a = *random action*
4: **Else if** $p \geq \varepsilon_k$
5: $a = \underset{a}{argmax}Q(s, a)$
6: **End**

算法。自适应动作选择策略可以表示为

$$\varepsilon_k = \varepsilon_f + \frac{\varepsilon_i - \varepsilon_f}{1 + e^{\frac{k}{\varepsilon_d}}} \tag{15}$$

其中ε_k表示第k集贪婪因子，ε_i表示初始贪婪因子，$\varepsilon_{\mathbf{f}}$表示最终贪婪因子，ε_d表示ε贪婪因子的衰减率。第 k 集的自适应 ε 贪婪策略的伪代码在算法 1 中给出。

3) Path Smoothing: DDQN算法的输出动作是离散的，而AMR在实际场景中的转向操作是连续的。为了更好地使规划路径更符合AMR的实际要求，需要使用贝塞尔曲线来平滑规划路径，因为它具有编程难度低、计算成本低、轨迹连续性强等优点[45]，[46]。因此，DDQN算法框架中引入贝塞尔曲线来实现端到端的平滑路径规划。从一阶到三阶贝塞尔曲线的计算表达式为

$$p_i^k = \begin{cases} p_i^0, & i = 0, 1, 2, \ldots, n-k \\ (1-t)p_i^{k-1} + tp_{i+1}^{k-1}, & i = 0, 1, 2, \ldots, n-k \end{cases} \tag{16}$$

其中，p_i^0 表示第 i 个初始控制点，p_i^k 表示第 i 个 K 阶控制点（k=1,2...,n），t 表示比例系数。

根据(16)式，n阶控制点的计算公式总是包含两个n-1阶控制点，因此最终可以根据初始控制点计算出n阶控制点的计算公式。二阶贝塞尔曲线可以保证路径的平滑性，并且比高阶贝塞尔曲线具有更快的计算速度。因此，串联多条二阶贝塞尔曲线来平滑路径的转折点，计算表达式为

$$B(t) = \left(1 - t^2\right)p_i^0 + 2(1-t)p_{i+1}^0 + t^2 p_{i+2}^0, t \in [0, 1] \tag{17}$$

其中p_i^0、p_{i+1}^0和p_{i+2}^0表示三个连续的初始化控制点。

4) AMR Path-Planning Algorithm Process: 综上所述，IDDQN算法应用于AMR路径规划的流程图如图10所示。该算法的伪代码如算法2所示，图11中连接线上的数字对应于算法2的流程。算法2的流程大致如下所示。

Step 1: 首先设置训练参数，主要包括状态空间、动作集、初始学习率等；

算法2 基于IDDQN的AMR路径规划算法1：输入：交互次数N；动作集A；学习率α；折扣系数 γ；体验重播池最大大小D；小批量M；当前神经网络值Q；目标神经网络值Q'。2：初始化网络参数θ和θ'，令$\theta' = \theta$，并初始化重放池R。 3：设置最大步数n_{max}。 4：对于第 1 集=，执行

5：初始化环境空间并获取起始状态$s = (x_0, y_0)$。

6：对于步骤=1，执行7：S为当前状态的输入，获取当前状态对应的所有Q值输出，并使用自适应ε贪婪策略来选择Q值对应的动作。设置 $a_t =$
$$\begin{cases} \underset{a'}{arg\ max}Q(s, a; \theta)\ with\ probability\ 1 - \varepsilon \\ random\ action \quad otherwise \end{cases}$$
8：动作a由AMR执行，AMR与外部环境交互接收奖励r和下一个状态s'。 9：将(s, a, r, s')存储到经验回放池中。

10: Let $s = s'$.
11：从D开始的R过渡(s_i, a_i, r_i, s_i')的随机样本小批量，并计算目标Q值的y_i。
12: Set $y_i = \begin{cases} r_i \quad if \quad episode\ terminates\ at\ step\ i \\ r_i + \gamma Q'(S', \underset{a'}{arg\ max}Q(s', a'; \theta); \theta^-)\ otherwise \end{cases}$
13：计算损失函数$L(\theta) = E[(y_i - Q(s, a; \theta))^2]$，通过神经网络的梯度方向反向传播来更新参数$\theta$。 14：替换目标参数：$\theta^- = \tau\theta + (1-\tau)\theta^-$。 15：如果$s'$为目标状态，则当前交互结束，然后使用Bezier曲线平滑路径。否则进入第7步。

16：输出：最优路径向量集。 17: 结束
18: 结束

其次，建立当前神经网络和目标神经网络；第三，随机初始化参数θ和θ^-，以及$\theta = \theta^-$ (第2行)。

Step 2: AMR 探索环境的过程（第 5-8 行）。在每个episode开始时，都会初始化环境信息，并且在探索过程中，将当前状态输入到当前网络，并输出相关动作的Q值（第5行和第6行）。使用改进的自适应 ε 贪婪策略获取并执行操作 a（第 7 行）。 AMR 与环境交互以接收奖励 r 和接下来的状态 s (第 8 行)。

Step 3: 当前和目标神经网络的更新过程（第 9-14 行）。首先，将发现过程中获得的多样化样本数据保存在经验回放数据池中，然后从经验池中随机采样小批量的经验数据来更新当前网络（第9-11行）。目标 Q 值在（第 12 行）中更新。其次，基于神经网络梯度的反向传播，计算损失函数，并

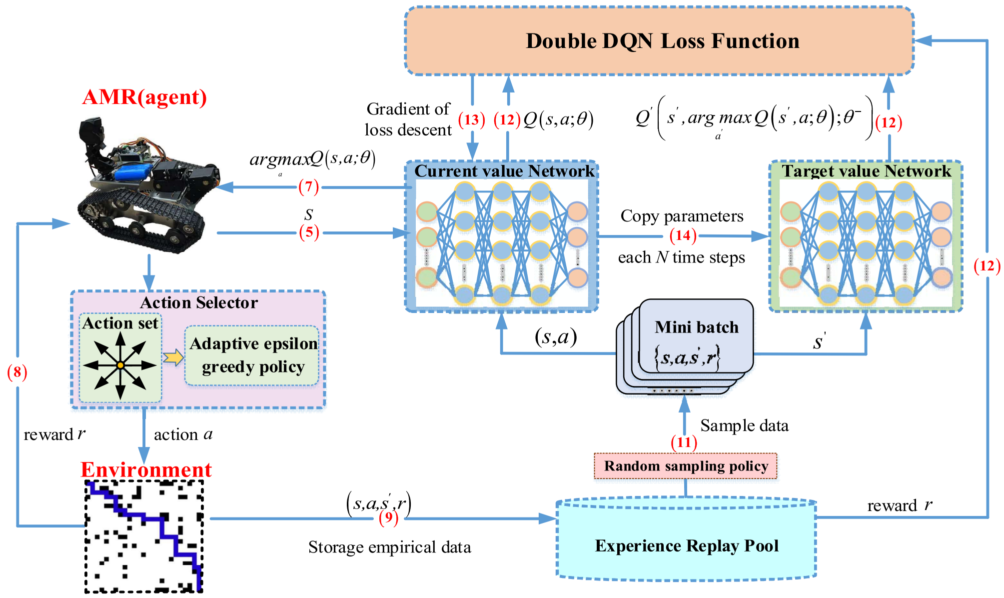

Fig. 11. Flowchart of the AMR's path-planning algorithm based on improved DDQN.

current value of parameter θ is updated (line 13). Third, the soft update of target network is performed (line 14).

Step 4: After multiple episodes, the final neural network parameters are saved, and AMR outputs the optimal path (line 15).

IV. Simulation Investigation

To verify the feasibility of the proposed IDDQN algorithm, the simulation model for the proposed IDDQN and the other four path-planning algorithms are established in the PyCharm platform, and four visual grid simulation environments with 30×30 dimension are designed by the gym framework. The hyperparameter configuration adopted by the IDDQN algorithm is summarized in Table I. The operating system device used for training the algorithm is Windows 11, the CPU is i7-12700H, and the GPU is NVIDIA RTX 3050 TI. In the same four unknown obstacle environments, the proposed IDDQN algorithm is compared with baseline algorithms to verify the effectiveness and superiority of the IDDQN algorithm. Meanwhile, the robustness of the IDDQN algorithm is initially assessed in unknown environments with random disturbances.

A. Baseline Algorithms

The algorithms for comparison are described as follows.

1) *IDDQN (Proposed):* Based on the DRL, action selection policy, DNNs, and reward function are jointly optimized. By introducing the comprehensive reward function, the path-planning security and search efficiency of AMR are improved.
2) *APF [10]:* The APF algorithm can quickly and flexibly plan an effective path by calculating the gradient of the potential field in the environment, and the path trajectory can maintain a safe distance from the obstacles.
3) *RRT [47]:* The RRT algorithm can explore the solution space by continuously expanding a randomly generated tree. It can adapt to various environments and avoid local optimal solutions while achieving efficient real-time path planning.
4) *A* [48]:* The A* algorithm can precisely guide robots in environmental exploration by leveraging both breadth-first search and best first search principles, and ensures finding the shortest path within a finite state space.

TABLE I
Hyperparameters Setting

Name of parameters	Value
Action space size	8
Learning rate α	0.0025
Decay factor γ	0.9
Star ε-greedy value	0.9
Final ε-greedy value	0.01
Exploration decay rate	500
Experience Replay pool	10000
Mini-batch size	64
Target network update step	4
Hidden layers size	2
Number of neurons	128
Activation function	ReLu
$(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$	(1, -0.35, -1, -1)

According to the advantages of APF algorithm, RRT algorithm and A* algorithm, the performances of the path results planned by different algorithms are assessed in four aspects: 1) path length; 2) path smoothness; 3) the distance between path trajectory and obstacles; and 4) the algorithm inference time (IT), so as to evaluate the superiority and effectiveness of the proposed IDDQN algorithm.

Additionally, the computational complexity of all comparation algorithms is listed below. The computational complexity of the proposed IDDQN algorithm is $O(EP*N)$, which is similar to the time complexity of the traditional DQN algorithm, where N is the number of time steps of each episode, and EP is the number of episodes. The time complexity of the

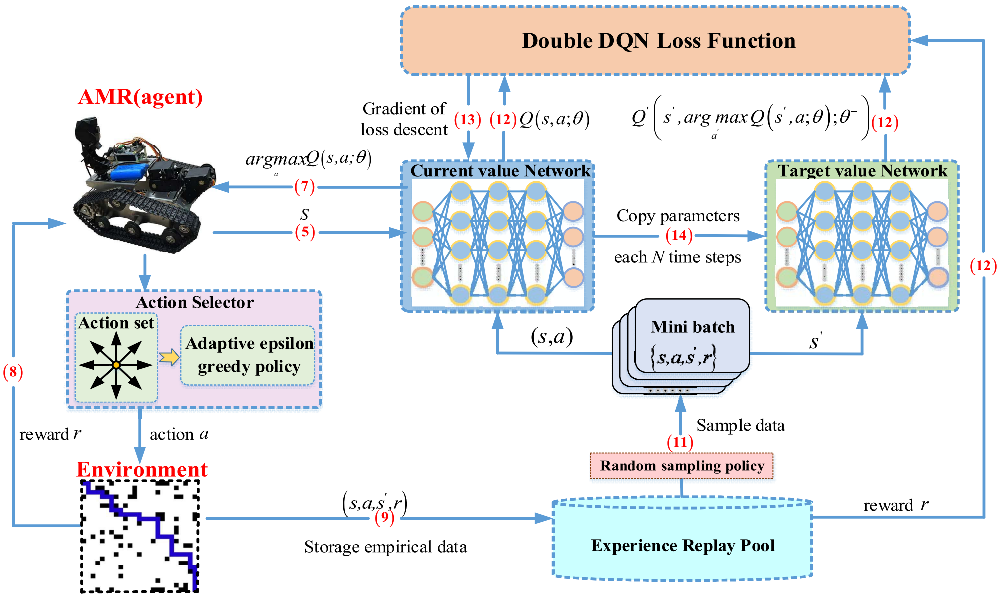

图11.基于改进DDQN的AMR路径规划算法流程图。

参数 θ 的当前值已更新（第 13 行）。第三，执行目标网络的软更新（第14行）。

Step 4: 经过多次训练后，最终的神经网络参数被保存，AMR 输出最优路径（第 15 行）。

四. 模拟研究

为了验证所提出的IDDQN算法的可行性，在PyCharm平台上建立了所提出的IDDQN和其他四种路径规划算法的仿真模型，并通过gym框架设计了四个30×30维度的可视化网格仿真环境。 IDDQN算法采用的超参数配置总结在表1中。用于训练算法的操作系统设备是Windows 11，CPU是i7-12700H，GPU是NVIDIA RTX 3050 TI。在相同的四种未知障碍物环境中，将所提出的IDDQN算法与基线算法进行比较，验证IDDQN算法的有效性和优越性。同时，初步评估了IDDQN算法在具有随机扰动的未知环境中的鲁棒性。

A. Baseline Algorithms

比较算法描述如下。

1) *IDDQN (Proposed):* 基于DRL，对动作选择策略、DNN和奖励函数进行联合优化。通过引入综合奖励函数，提高了AMR的路径规划安全性和搜索效率。

2) *APF [10]:* APF算法通过计算环境中势场的梯度可以快速灵活地规划有效路径，并且路径轨迹可以与障碍物保持安全距离。

3) *RRT [47]:* RRT算法可以通过不断扩展随机生成的解空间来探索解空间树。它可以适应各种环境，避免局部最优解，同时实现高效的实时路径规划。

4) *A* [48]:* A*算法可以利用广度优先搜索和最佳优先搜索原则精确引导机器人进行环境探索，并确保在有限状态空间内找到最短路径。

表1 超参数设置

Name of parameters	Value
Action space size	8
Learning rate α	0.0025
Decay factor γ	0.9
Star ε-greedy value	0.9
Final ε-greedy value	0.01
Exploration decay rate	500
Experience Replay pool	10000
Mini-batch size	64
Target network update step	4
Hidden layers size	2
Number of neurons	128
Activation function	ReLu
$(\lambda_1,\lambda_2,\lambda_3,\lambda_4)$	(1, -0.35, -1, -1)

根据APF算法、RRT算法和A*算法的优点，从四个方面评估不同算法规划的路径结果的性能：1）路径长度； 2）路径平滑度； 3）路径轨迹与障碍物之间的距离； 4）算法推理时间（IT），以评估所提出的IDDQN算法的优越性和有效性。

此外，下面列出了所有比较算法的计算复杂度。所提出的IDDQN算法的计算复杂度为$O(EP*N)$，与传统DQN算法的时间复杂度相似，其中N是每个episode的时间步，EP是episode的数量。的时间复杂度为

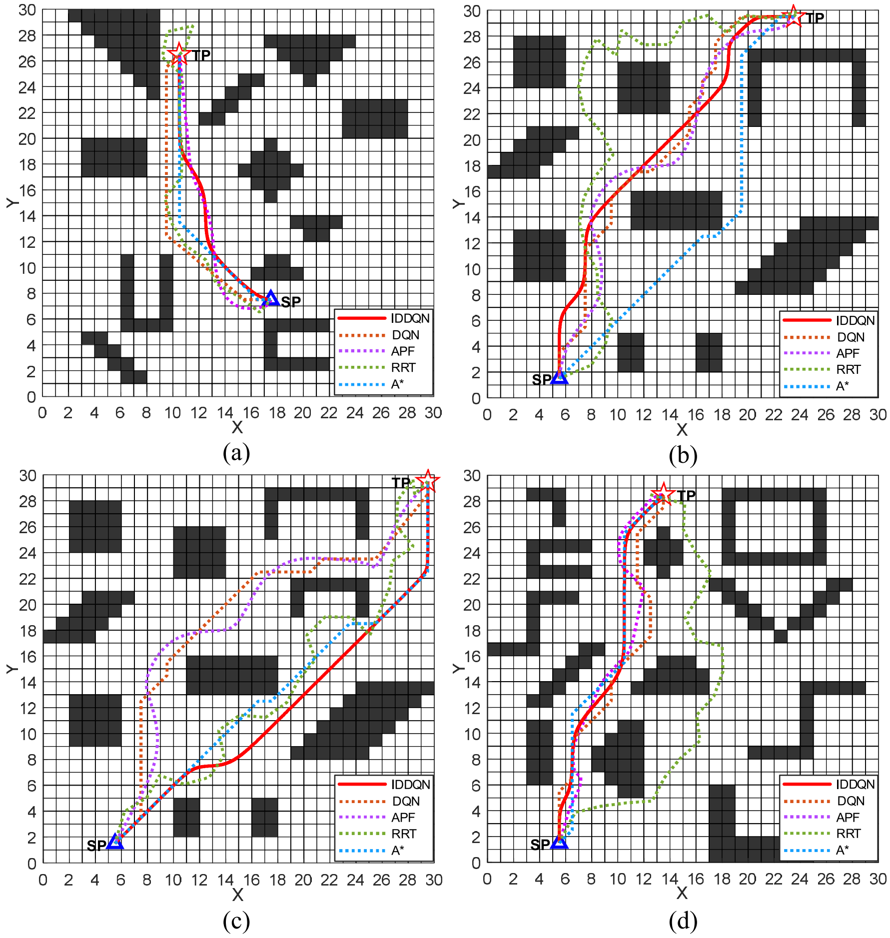

Fig. 12. Simulation results of different path-planning methods.

A* algorithm can be expressed as $O(N_1 \log N_1)$, wherein N_1 denotes the number of nodes. The time complexity expression of the APF algorithm is $O(N_2 * M)$, where N_2 denotes the number of iterations, and M denotes the number of obstacles. Besides, the time complexity of the RRT algorithm is $O(N_3^2)$, where N_3 denotes the vertex number of the tree.

B. Simulations Under Unknown Environment Without Disturbance

Here, four types of unknown environments with different terrain features and obstacle densities are designed to evaluate the performance of the proposed IDDQN algorithm. First, five key performance indicators (KPIs) are established to assess the performance of various path-planning algorithms. The KPIs include the number of path corners (PCs), maximum path turning angle (MPTA), minimum collision to distance (MDTC), average path length (APL), as well as algorithm IT. The planning paths of those five algorithms under four different simulation environments are shown in Fig. 12, and the KPIs values are shown in Table II.

As shown in Fig. 12, all the five path-planning algorithms can successfully generate collision-free paths in four different unknown environments. On the one hand, based on the trajectory planned by the three commonly used traditional path-planning methods like APF, RRT, and A*, and the comparative analysis of KPIs in Table II, it can be observed that the APF algorithm can compute AMR motion trajectory in real time under the premise of ensuring motion safety. However, the APF algorithm is susceptible to fall into local minima, which may lead to the presence of local oscillations and longer path lengths in the generated paths. The RRT algorithm demonstrates high search efficiency in path planning. Nevertheless, the paths generated by the RRT algorithm have longer path lengths, more path turning points, and the path trajectory is closer to the obstacles, which cannot ensure the safety of AMR path planning. Additionally, although the path planned by the A* algorithm is relatively shorter with fewer turning points, there is a risk of collision between the trajectory generated by A* algorithm and the edges of obstacles. Moreover, as the complexity of the environment increases, the IT of A* algorithm will significantly increase. On the other hand, by comparing the performance results of the DQN and the proposed IDDQN algorithm in path planning under four unknown complex environments, it can be concluded that the trajectories generated by the DQN and IDDQN algorithms can maintain the distance from obstacles at least 0.5 m, and can generally outperform three traditional path-planning algorithms. More importantly, compared with the DQN algorithm, the APL of the IDDQN algorithm is,

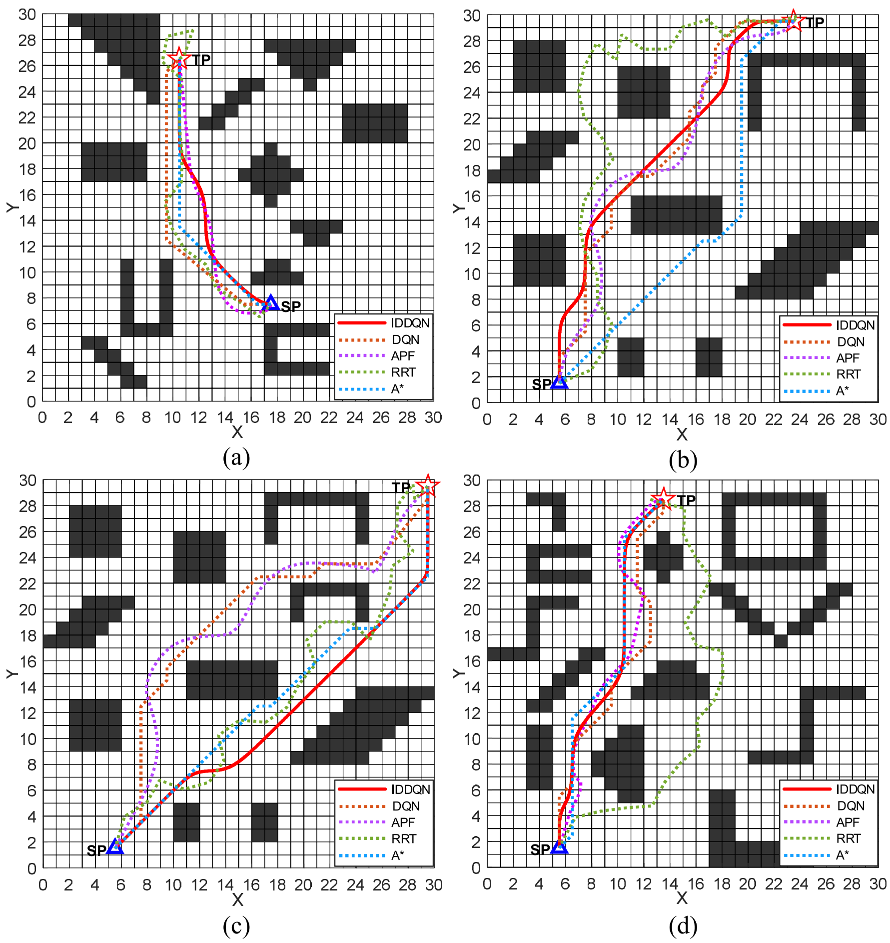

图 12. 不同路径规划方法的仿真结果。

A*算法可以表示为$O(N_1 \log N_1)$，其中N_1表示节点数量。 APF算法的时间复杂度表达式为$O(N_2 * M)$，其中N_2表示迭代次数，M表示障碍物数量。此外，RRT算法的时间复杂度为$O(N_3^2)$，其中N_3表示树的顶点数。

B. Simulations Under Unknown Environment Without Disturbance

这里，设计了四种具有不同地形特征和障碍物密度的未知环境来评估所提出的 IDDQN 算法的性能。首先，建立五个关键绩效指标（KPI）来评估各种路径规划算法的性能。 KPI 包括路径拐角数 (PC)、最大路径转向角度 (MPTA)、最小碰撞距离 (MDTC)、平均路径长度 (APL) 以及算法 IT。这五种算法在四种不同仿真环境下的规划路径如图12所示，KPI值如表2所示。

如图 12 所示，所有五种路径规划算法都可以在四种不同的未知环境中成功生成无碰撞路径。一方面，基于三种常用传统的轨迹规划

通过与APF、RRT、A*等路径规划方法的比较，以及表2中KPI的对比分析，可以看出APF算法可以在保证运动安全的前提下实时计算AMR运动轨迹。然而，APF算法容易陷入局部最小值，这可能导致生成的路径中存在局部振荡和较长的路径长度。 RRT 算法在路径规划中表现出较高的搜索效率。然而，RRT算法生成的路径路径长度较长、路径转折点较多、路径轨迹距离障碍物较近，无法保证AMR路径规划的安全性。另外，虽然A*算法规划的路径相对较短、转弯点较少，但A*算法生成的轨迹存在与障碍物边缘发生碰撞的风险。而且，随着环境复杂度的增加，A*算法的IT也会显着增加。另一方面，通过比较DQN和IDDQN算法在四种未知复杂环境下路径规划的性能结果，可以得出结论，DQN和IDDQN算法生成的轨迹可以保持与障碍物的距离至少0.5 m，并且总体上优于三种传统路径规划算法。更重要的是，与DQN算法相比，IDDQN算法的APL是，

TABLE II
COMPARISON OF KPIs AMONG DIFFERENT PATH-PLANNING ALGORITHMS

Environment No.	Algorithm name	KPIs				
		Average path length (m)	Number of path corners	Mini collision to obstacle (m)	Inference time (s)	Max corner degree (°)
Env. (a)	IDDQN	21.428	4	0.749	0.00094	15
	DQN	24.264	3	0.5	0.00104	45
	APF	23.406	6	1.1486	0.0582	83
	RRT	31.426	13	0.2434	0.0491	120
	A*	22.484	2	0	0.1305	45
Env. (b)	IDDQN	35.842	6	0.7494	0.00117	41
	DQN	39.799	14	0.7071	0.00121	45
	APF	66.369	9	1.4296	0.0630	62
	RRT	50.988	23	0.9096	0.0512	117
	A*	36.704	4	0	0.3725	45
Env. (c)	IDDQN	39.229	3	0.7071	0.00110	41
	DQN	42.300	9	0.5	0.00132	45
	APF	43.408	12	1.4467	0.0716	84
	RRT	48.678	24	0.0386	0.0622	139
	A*	39.742	5	0	0.6673	45
Env. (d)	IDDQN	28.248	5	0.7071	0.00147	21
	DQN	31.324	10	0.5	0.00151	45
	APF	31.559	11	1.4619	0.0733	60
	RRT	39.781	16	0.3742	0.0617	103
	A*	30.612	5	1.4142	0.6304	45

respectively, reduced by 11.69%, 8.49%, 7.26%, and 9.82% under the four complex environments, and the planning path generated by the IDDQN algorithm has better smoothness and fewer path turning points.

Fig. 13 reveals the variation of reward values during the training process of the DQN and IDDQN algorithms under four various environments. Due to the extensive exploration of the environment by agent (AMR) in the early training stage of our proposed IDDQN algorithm, the reward function curve of the IDDQN algorithm appears obvious oscillation. Subsequently, the AMR accelerates the learning process by leveraging the experiential knowledge gained from exploration, ultimately leading to the stabilization of the learned rewards. As the environmental complexity of the obstacles increases, both IDDQN and DQN need more exploration steps to make the reward value becoming stable. Although the DQN algorithm could achieve stability in the reward curve after a period of continuous learning and trial and error, its convergence speed is slow due to the overestimation of Q-values and the presence of excessive ineffective action-selection policies. Furthermore, the reward curve exhibits significant fluctuations in the later stage of training process. By comparing the trend of reward value change for proposed IDDQN and the DQN, it can be seen that the number of iteration steps required by the IDDQN algorithm to achieve stable reward value is reduced about by 27.31%, 32.17%, 52.91%, and 26.40% in compared with the DQN algorithm under four different environments. The comparison results show that the IDDQN has higher learning efficiency, global search ability, and excellent convergence.

The radar chart shown in Fig. 14 is generated based on the data recorded in Table II, which facilitate the analysis of the difference in path-planning performance between the IDDQN algorithm and the other four algorithms. From Fig. 14, it can be seen that the inference speed of the IDDQN algorithm model is much better than APF, RRT, and A* algorithms

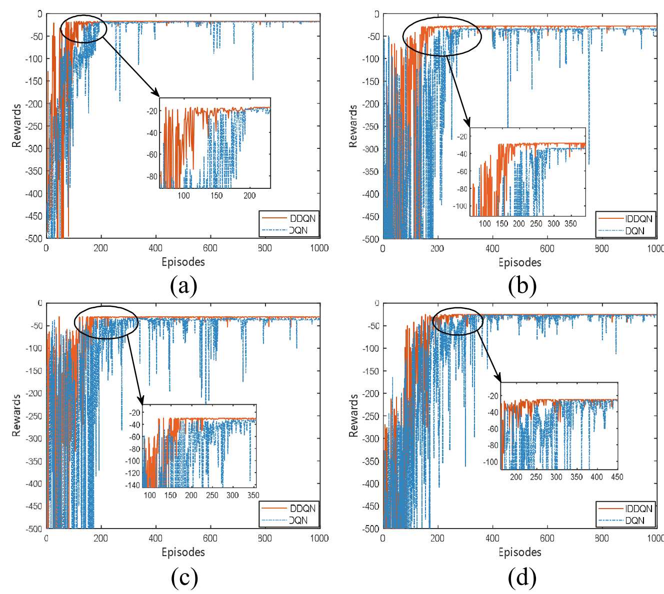

Fig. 13. Changes in the reward values of IDDQN and DQN during the training process.

under the four experimental environments. The path length, the number of turn points, and the maximum turning angle of the path generated by the IDDQN algorithm are smaller than DQN algorithm and other traditional methods. Therefore, the IDDQN algorithm adopt the end-to-end training mode, which has better integrated path-planning capability.

C. Simulations Under Unknown Environment With Disturbance

AMR may experience changes in external environment factors during movement, such as insufficient road friction and nonuniform ground conditions, which may lead to the poor stability and the directional slip for the AMR. Therefore, the

表2 不同路径规划算法的KPI比较

Environment No.	Algorithm name	KPIs				
		Average path length (m)	Number of path corners	Mini collision to obstacle (m)	Inference time (s)	Max corner degree (°)
Env. (a)	IDDQN	21.428	4	0.749	0.00094	15
	DQN	24.264	3	0.5	0.00104	45
	APF	23.406	6	1.1486	0.0582	83
	RRT	31.426	13	0.2434	0.0491	120
	A*	22.484	2	0	0.1305	45
Env. (b)	IDDQN	35.842	6	0.7494	0.00117	41
	DQN	39.799	14	0.7071	0.00121	45
	APF	66.369	9	1.4296	0.0630	62
	RRT	50.988	23	0.9096	0.0512	117
	A*	36.704	4	0	0.3725	45
Env. (c)	IDDQN	39.229	3	0.7071	0.00110	41
	DQN	42.300	9	0.5	0.00132	45
	APF	43.408	12	1.4467	0.0716	84
	RRT	48.678	24	0.0386	0.0622	139
	A*	39.742	5	0	0.6673	45
Env. (d)	IDDQN	28.248	5	0.7071	0.00147	21
	DQN	31.324	10	0.5	0.00151	45
	APF	31.559	11	1.4619	0.0733	60
	RRT	39.781	16	0.3742	0.0617	103
	A*	30.612	5	1.4142	0.6304	45

在四种复杂环境下分别降低了11.69%、8.49%、7.26%和9.82%，IDDQN算法生成的规划路径具有更好的平滑性和更少的路径拐点。

图13揭示了DQN和IDDQN算法在四种不同环境下训练过程中奖励值的变化。由于在我们提出的IDDQN算法的早期训练阶段，代理（AMR）对环境进行了广泛的探索，IDDQN算法的奖励函数曲线出现了明显的振荡。随后，AMR 通过利用从探索中获得的经验知识来加速学习过程，最终实现学习奖励的稳定。随着障碍物环境复杂性的增加，IDDQN和DQN都需要更多的探索步骤才能使奖励值变得稳定。尽管DQN算法经过一段时间的不断学习和试错后可以实现奖励曲线的稳定性，但由于对Q值的高估以及存在过多无效的动作选择策略，其收敛速度较慢。此外，奖励曲线在训练过程的后期表现出显着的波动。通过比较所提出的IDDQN和DQN的奖励值变化趋势，可以看出，在四种不同环境下，IDDQN算法获得稳定奖励值所需的迭代步数比DQN算法减少了约27.31%、32.17%、52.91%和26.40%。比较结果表明IDDQN具有更高的学习效率、全局搜索能力和良好的收敛性。

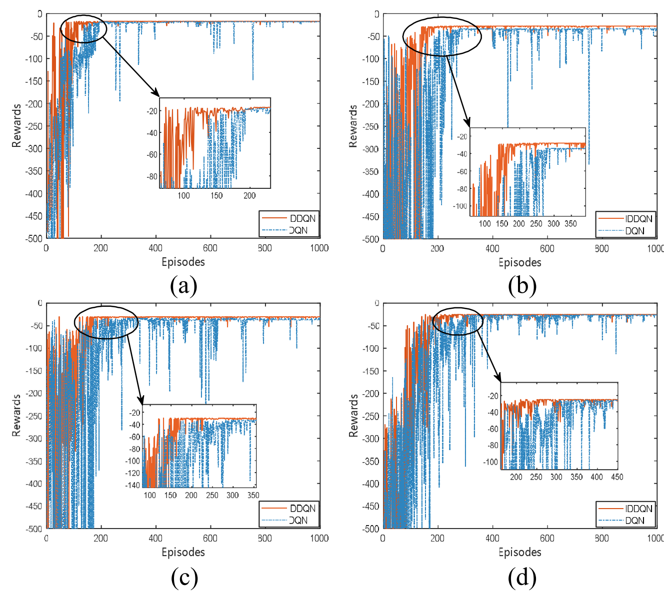

图13.训练过程中IDDQN和DQN奖励值的变化。

图14所示的雷达图是根据表II记录的数据生成的，有助于分析IDDQN算法与其他四种算法之间路径规划性能的差异。从图14可以看出，IDDQN算法模型的推理速度远优于APF、RRT、A*算法

四种实验环境下。 IDDQN算法生成的路径的路径长度、转弯点数、最大转弯角度均小于DQN算法等传统方法。因此，IDDQN算法采用端到端的训练模式，具有更好的综合路径规划能力。

C. Simulations Under Unknown Environment With Disturbance

AMR在运动过程中可能会受到外界环境因素的变化，如路面摩擦力不足、地面条件不均匀等，从而导致AMR稳定性差，出现定向滑移。因此，

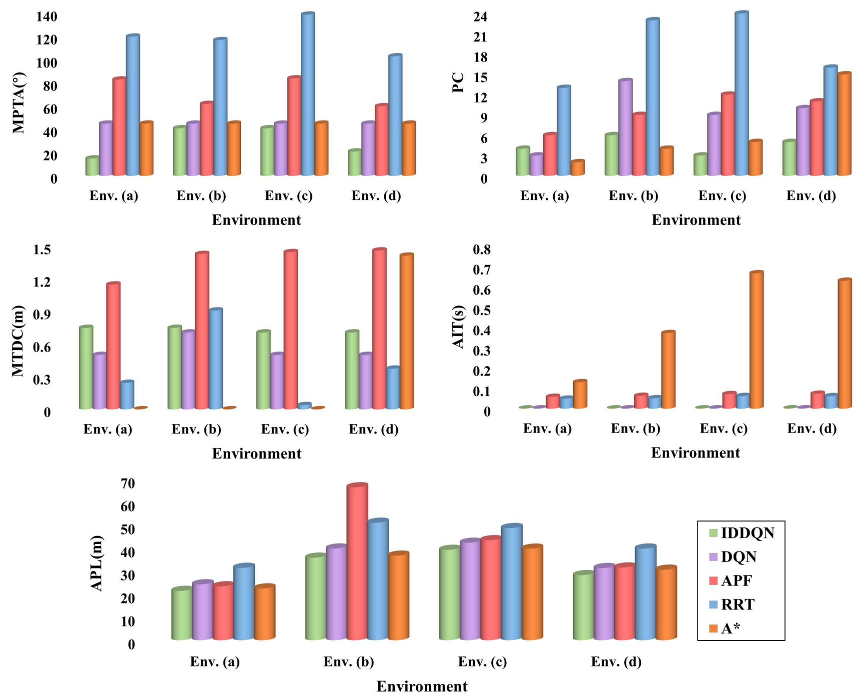

Fig. 14. Comparison results of KPIs among different path-planning algorithms in four environments.

AMR's motions will be determined by the action selection policy and the external environment information. Specifically, the influence of displacement disturbance on AMR path planning can be described as: AMR will generate random offset (a, b) in the x-axis or y-axis direction after passing through the position of random disturbance $(x, y)_{\text{current}}$ and arriving at the next state $(x, y)_{\text{next}}$, wherein $(x, y)_{\text{next}}$ can be described as $(x, y)_{\text{next}} = (x, y)_{\text{current}} + (a, b)$.

To assess the adaptability and robustness of the proposed IDDQN algorithm in unknown environments with random disturbances, multiple types of disturbances are randomly set in two single-target complex obstacle environments of size 20×20 and two multitarget complex obstacle environments of size 30×30. The location and size of random disturbances in the single-target environments are recorded in Table III. In the multiobjective environments, the AMR's path-planning process is divided into different phases by each target area, and the location and size of random disturbances in each operational phase of AMR are recorded in Table IV. The path-planning results of the IDDQN algorithm in the training environment with and without disturbance are shown in Fig. 15.

Meanwhile, Fig. 16 presents five KPIs to appraise the influence of disturbances in the environment on the path-planning results. These KPIs include the APL, total exploration steps required to reach the stabilizing reward value (TS), minimum distance to collision (MDTC), the number of PCs, and MPTA.

It can be observed from Fig. 15 that the IDDQN algorithm enables the AMR to avoid obstacles and reach each target area regardless of the presence of disturbances in the single-target environments or the multitarget environments.

TABLE III
RANDOM DISTURBANCE DISTRIBUTION AND SIZE IN SINGLE-TARGET OBSTACLE ENVIRONMENTS

Environments	Disturbances	*(a, b)*
Single target (a)	x=5.5	(0,1)
	y=12.5	(0,1)
Single target (b)	x=5.5	(0,1)
	y=9.5	(1,0)

TABLE IV
RANDOM DISTURBANCE DISTRIBUTION AND SIZE IN MULTIPLE TARGET OBSTACLE ENVIRONMENTS

Environments	Phases	Disturbances	*(a, b)*
Multiple continuous targets (b)	Phase 1	x=8.5	(0,1)
	Phase 2	y=14.5	(1,0)
	Phase 3	x=18.5	(0,1)
Multiple continuous targets (c)	Phase 1	x=9.5	(0,1)
	Phase 2	x=12.5	(-1,0)
	Phase 3	y=23.5	(-1,0)
	Phase 4	x=18.5	(0,1)

However, the presence of environmental disturbances has caused deviations in the paths planned by the IDDON algorithm compared to those planned in undisturbed environments. Moreover, the more disturbances information and continuous TPs in the environment, the more significant the path deviations become. In the environmental disturbances, the final path generated by the IDDON is longer and contains more path turning points compared to the path planning without disturbance.

By comparing the KPIs data in Fig. 15, it can be found that the path length and total exploration steps by the IDDQN algorithm are significantly influenced by the environmental

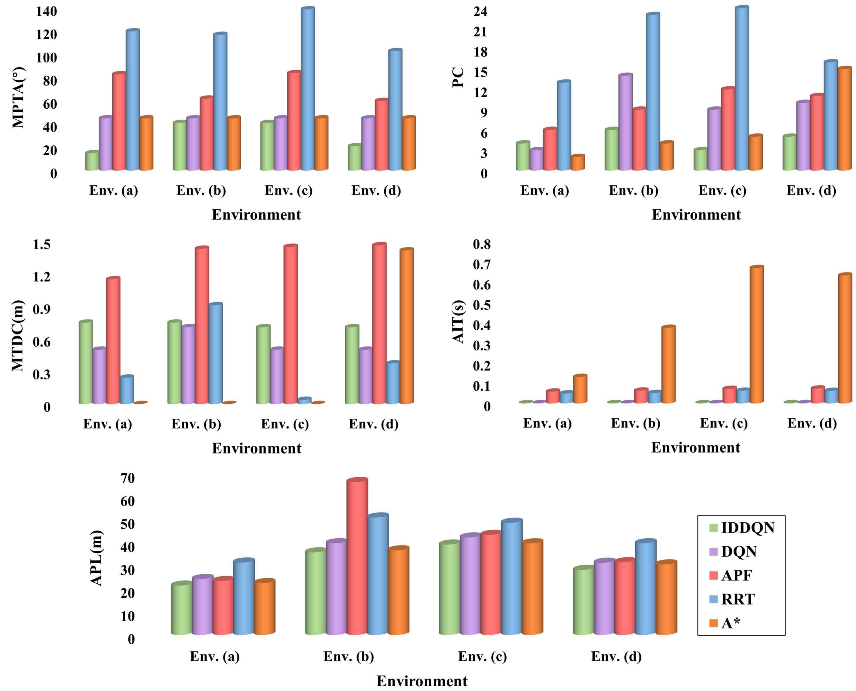

图14.四种环境下不同路径规划算法的KPI比较结果。

AMR的动议将由行动选择政策和外部环境信息决定。具体来说，位移扰动对AMR路径规划的影响可以描述为：AMR经过随机扰动$(x, y)_{current}$的位置，到达下一个状态$(x, y)_{next}$后，会在x轴或y轴方向产生随机偏移(a, b)，其中$(x, y)_{next}$可以描述为$(x, y)_{next} = (x, y)_{current} + (a, b)$。

为了评估所提出的IDDQN算法在具有随机扰动的未知环境中的适应性和鲁棒性，在两个尺寸为20×20的单目标复杂障碍物环境和两个尺寸为30×30的多目标复杂障碍物环境中随机设置多种类型的扰动。单目标环境中随机扰动的位置和大小记录在表III中。在多目标环境中，AMR的路径规划过程按每个目标区域分为不同的阶段，AMR每个运行阶段的随机扰动的位置和大小记录在表IV中。 IDDQN算法在有干扰和无干扰的训练环境下的路径规划结果如图15所示。

同时，图16提出了五个KPI来评估环境干扰对路径规划结果的影响。这些 KPI 包括 APL、达到稳定奖励值 (TS) 所需的总探索步骤、最小碰撞距离 (MDTC)、PC 数量和 MPTA。

从图15可以看出，IDDQN算法使得AMR能够避开障碍物并到达每个目标区域，无论单目标环境或多目标环境中是否存在干扰。

表 III 单目标障碍物环境中的随机扰动分布和大小

Environments	Disturbances	(a, b)
Single target (a)	x=5.5	(0,1)
	y=12.5	(0,1)
Single target (b)	x=5.5	(0,1)
	y=9.5	(1,0)

表 IV 多目标障碍物环境中的随机扰动分布和大小

Environments	Phases	Disturbances	(a, b)
Multiple continuous targets (b)	Phase 1	x=8.5	(0,1)
	Phase 2	y=14.5	(1,0)
	Phase 3	x=18.5	(0,1)
Multiple continuous targets (c)	Phase 1	x=9.5	(0,1)
	Phase 2	x=12.5	(-1,0)
	Phase 3	y=23.5	(-1,0)
	Phase 4	x=18.5	(0,1)

然而，环境干扰的存在导致IDDON算法规划的路径与未干扰环境中规划的路径存在偏差。此外，环境中的干扰信息和连续TP越多，路径偏差就越显着。在环境干扰下，与无干扰的路径规划相比，IDDON生成的最终路径更长并且包含更多的路径转折点。

通过比较图15中的KPI数据，可以发现IDDQN算法的路径长度和总探索步数受环境影响显着。

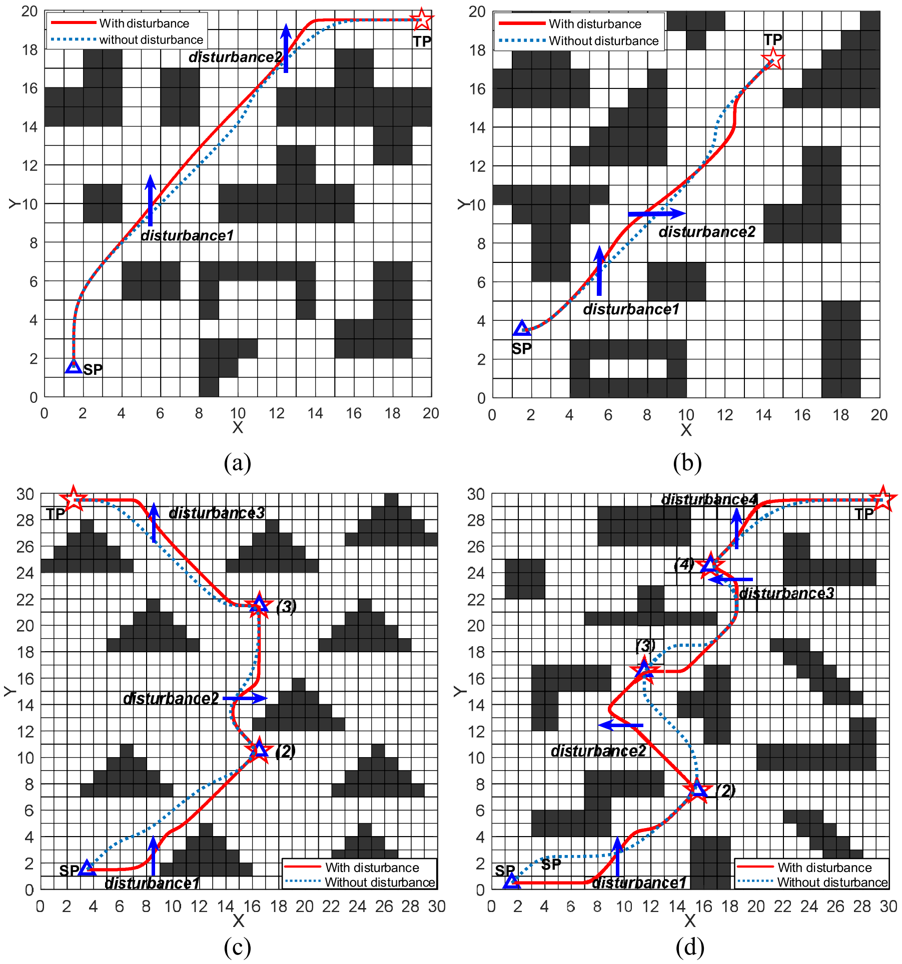

Fig. 15. (a) and (b) AMR's trajectory in the single-target unknown disturbance environment. (c) and (d) AMR's trajectory in the multitarget unknown disturbance environment.

disturbances in two different environments. Especially, according to Fig. 15(d), the path length obtained under unknown environment with disturbance is increased by 10.12% compared to the original path, and the number of exploration steps that is required to reach a stable reward value for the IDDQN algorithm is increased by 66.06%. Additionally, the IDDQN algorithm needs to mitigate the impact of random disturbances in the environment while avoiding obstacles, resulting in larger turning angles and more path turning points in the generated paths. Simulation experiments demonstrate that the random disturbances in the environment do not hinder the IDDQN algorithm from planning optimal paths for the AMR to reach all target areas, while ensuring that the path trajectories maintain a minimum safety distance from different obstacles. However, the presence of disturbances in the environment can increase the computational demands of the IDDQN algorithm. In summary, the IDDQN algorithm will continuously learn the disturbance information from scratch in unknown environments and can exhibit a certain level of adaptability and robustness. The proposed IDDQN algorithm model can be extended to apply for more environments, which provides a new solution for AMR path planning.

V. Conclusion

This article proposes an improved DDQN algorithm to deal with the AMR path-planning issue in comprehensive unknown environments. Initially, following establishment of the AMR path-planning MDP model, a comprehensive reward function is designed to quickly navigate the AMR to the target area while ensuring AMR to maintain a safe distance from the obstacles. Then, in the DDQN algorithm framework, it integrates the adaptive ε-greedy policy and the optimized DNN, which improves the global search capability and the convergence speed of the DDQN algorithm. Additionally, the Bezier curve algorithm is introduced to smooth the discrete actions output by the DDQN. Finally, the IDDQN algorithm is verified under varies comprehensive unknown environments. In four groups unknown environments, compared with DQN, A*, RRT, and APF algorithms, the paths planned by the proposed IDDQN algorithm is not only safer and shorter path for the AMR but also its inference speed is much better than traditional methods. Meanwhile, through the evaluation of the IDDQN algorithm's performance in unknown environments with random disturbance, both in single-target and multitarget scenarios, it is further demonstrated that the IDDQN algorithm exhibits better adaptability and robustness.

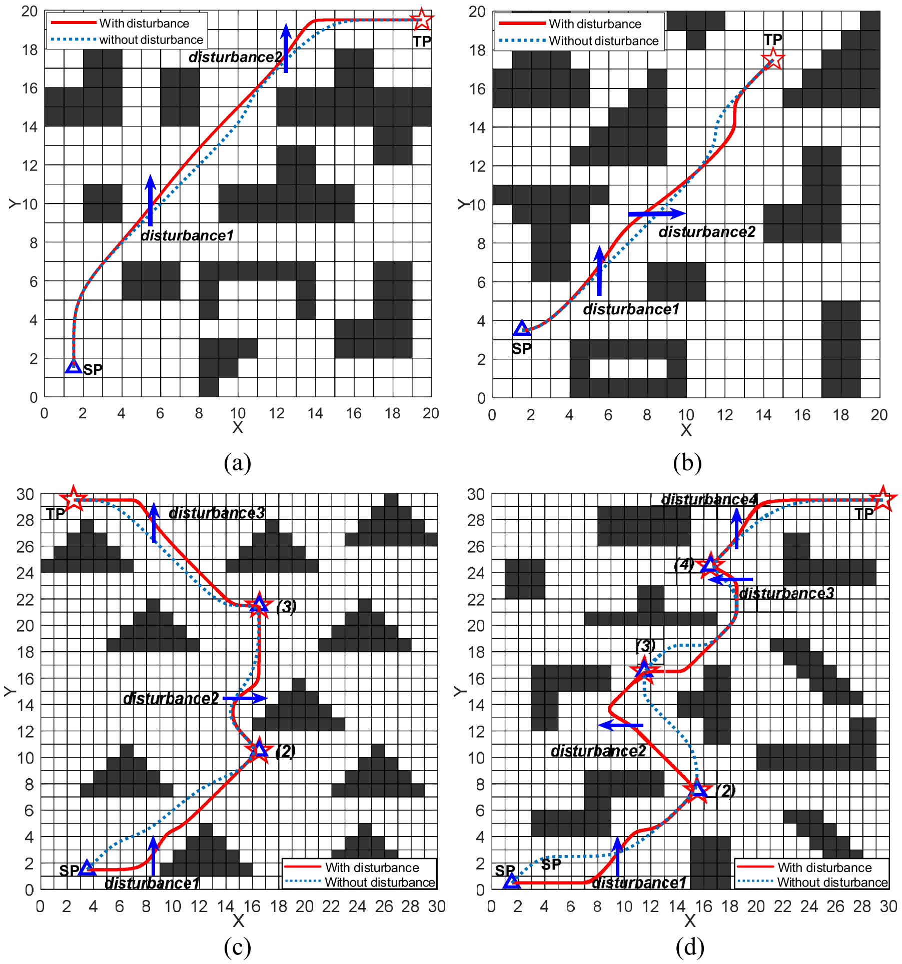

图1　5. (a)和(b)单目标未知扰动环境下AMR的轨迹。 (c) 和 (d) AMR 在多目标未知中的轨迹
干扰 平衡环境。

两个不同环境中的干扰。特别是，根据图15（d），在未知干扰环境下获得的路径长度比原始路径增加了10.12%，IDDQN算法达到稳定奖励值所需的探索步骤数增加了66.06%。另外，IDDQN算法需要在避开障碍物的同时减轻环境中随机扰动的影响，导致生成的路径具有更大的转弯角度和更多的路径转折点。仿真实验表明，环境中的随机扰动并不妨碍IDDQN算法规划AMR到达所有目标区域的最佳路径，同时保证路径轨迹与不同障碍物保持最小安全距离。然而，环境中存在的干扰会增加 IDDQN 算法的计算需求。综上所述，IDDQN算法会在未知环境中从头开始不断学习扰动信息，能够表现出一定程度的适应性和鲁棒性。所提出的IDDQN算法模型可以扩展以适用于更多环境，为AMR路径规划提供了新的解决方案。

五、结论

本文提出一种改进的DDQN算法来处理综合未知环境下的AMR路径规划问题。最初，在建立AMR路径规划MDP模型后，设计了综合奖励函数，以将AMR快速导航到目标区域，同时确保AMR与障碍物保持安全距离。然后，在DDQN算法框架中，融合了自适应ε-贪婪策略和优化的DNN，提高了DDQN算法的全局搜索能力和收敛速度。此外，还引入了贝塞尔曲线算法来平滑 DDQN 输出的离散动作。最后，在各种综合未知环境下对IDDQN算法进行了验证。在四组未知环境中，与DQN、A*、RRT和APF算法相比，所提出的IDDQN算法规划的路径不仅对于AMR来说更安全、更短，而且其推理速度也远优于传统方法。同时，通过评估IDDQN算法在随机干扰的未知环境下，无论是单目标还是多目标场景，进一步证明IDDQN算法表现出更好的适应性和鲁棒性。

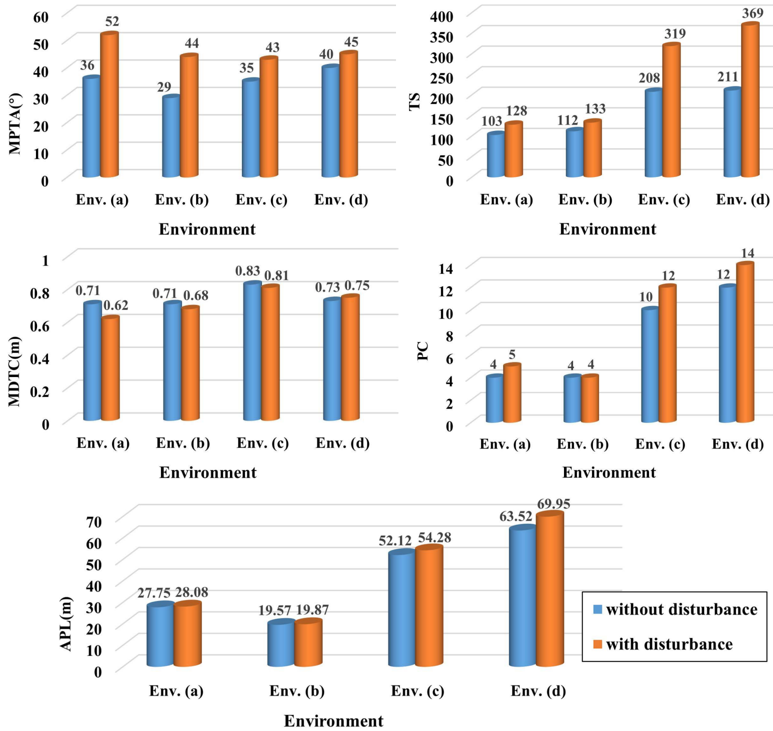

Fig. 16. Effects of disturbances on AMR path planning in different environments.

For future work, we will focus on sensor-based dynamic path planning, and the practical algorithm validation in hardware-in-loop experiment will be performed.

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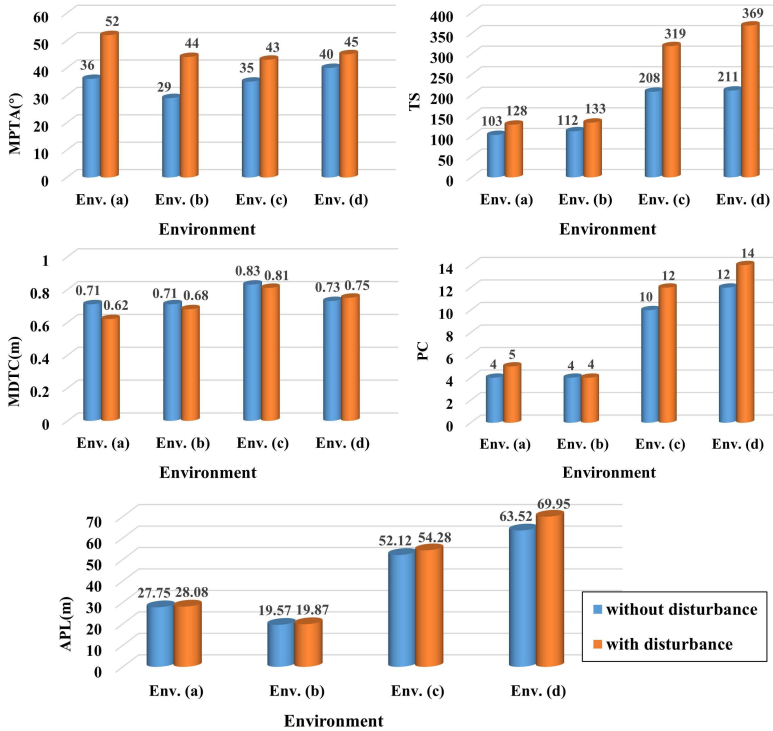

图 16. 不同环境下干扰对 AMR 路径规划的影响。

在未来的工作中，我们将重点研究基于传感器的动态路径规划，并在硬件在环实验中进行实际算法验证。

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